Borrowing Strength Without Borrowing Data

Private, Serverless and Tuning-Free Collaboration with Graph Hierarchical Borrowing

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Most clinical prediction tasks are small at the hospital that needs the model, while the data that would make them large sit in other hospitals that cannot share records and have no centre they all trust. We propose Graph Hierarchical Borrowing (GHB): each hospital shrinks its own model head toward a leave-one-out estimate of the network, by an amount its own data decide. In a Gaussian random-effects model, every hospital gains almost the same number of "virtual patients", $\sigma^2/(\tau^2 + M)$, so the smallest gain the most; privacy enters only through the error M of the network estimate; and computing it by gossip costs no privacy. For realistic heads, the network releases two privatized sums, of every patient's score and information, by secure summation without a server; each hospital turns them locally into a prior with the privacy noise propagated and integrates the heterogeneity out against its own evidence. Nothing is tuned. On eight cancer networks held out from its development, five from The Cancer Genome Atlas and three from the Clinical Proteomic Tumor Analysis Consortium, GHB reduced held-out log-loss to 0.916 of local estimation at $\epsilon = 1$ and 0.907 at $\epsilon = 10$, better than a hospital alone on all eight networks at both budgets (Holm-corrected $p = 0.016$). Against six federated and personalized methods tuned on twelve other networks it was more accurate on 7 of the 8 networks at $\epsilon = 1$ (0.916 against 0.950 for the strongest), a lead eight networks do not establish ($p = 0.109$), and level with them at $\epsilon = 10$ (4 of 8). Stress tests on the TCGA networks show that GHB's lead there survives fewer hospitals and larger heads but comes from fitting related networks together; a single cancer network fitted alone gives GHB little more than a well-regularized hospital. On the FLamby benchmark and on Camelyon17 histopathology it is competitive, leading clearly in one setting and by small margins or not at all in the others. Its gain over a hospital alone also holds across 19 real data sets, ten of which informed its design. At $\epsilon = 1$, no attacker can tell whether a patient's record was used with more than 8.1% advantage.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Applied computing** → **Health informatics**; • **Security and privacy** → **Privacy protections**.

Additional Key Words and Phrases: small data, multi-site learning, decentralized learning, gossip, differential privacy, empirical Bayes, hierarchical models, personalization, data silos, healthcare

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1 INTRODUCTION

Most clinical learning problems are small where the model is needed. More than six thousand rare diseases affect 3.5% to 5.9% of the population, mostly in cohorts of tens or hundreds of patients [Nguengang Wakap et al. 2020; Schaefer et al. 2020]; prediction models in common conditions are

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routinely built on a few dozen outcome events [Peduzzi et al. 1996; Riley et al. 2019]; almost none of the hundreds of COVID-19 models built from small local samples was fit for clinical use [Roberts et al. 2021; Wynants et al. 2020]; and deployed systems degrade at the point of use [Wong et al. 2021; Zech et al. 2018]. Healthcare is *collectively big and locally small* [Simões et al. 2026]: the data that would close the gap exist, but in other hospitals.

Mesothelioma shows the problem in miniature. In The Cancer Genome Atlas [Cancer Genome Atlas Research Network et al. 2013], the 9 institutions that contributed eligible patients hold 5 to 14 patients each, 72 in all. None can fit even an eight-coefficient model of two-year mortality on its own patients with any confidence; together they hold several times more information, and the same institutions treat other cancers, so what one disease teaches about age or stage can inform another. The question is how such hospitals can learn from each other's patients without seeing them.

Those hospitals cannot simply pool. Training on another hospital's records is secondary use, which needs its own legal basis and often excludes exporting patient-level data; the agreements that allow it are bilateral, slow and partial, so they form a sparse network, and often no institution is trusted or funded to be its centre [Cohen and Mello 2018; European Parliament and Council of the European Union 2016, 2025; Price and Cohen 2019]. The smallest hospitals, which need collaboration most, are also the easiest to re-identify [Rocher et al. 2019]. Existing methods each give up something this setting cannot spare. Federated learning needs a trusted server and fits one model [McMahan et al. 2017]; personalized variants fit one model per site but keep the server and must tune how much each site borrows [Li et al. 2021a]; decentralized learning removes the server but, in healthcare, lacks formal privacy or shares full models [Kalra et al. 2023; Warnat-Herresthal et al. 2021]; and private versions of all of them spend the budget over many rounds, with noise largest at the smallest sites [Bagdasaryan et al. 2019; McMahan et al. 2018]. Meta-analysis is exact and auditable, but estimates a single population model [DerSimonian and Laird 1986].

Graph Hierarchical Borrowing (GHB) starts from an old idea, shrinking each unit's estimate toward what the others suggest by an amount set by how much the units differ [Efron and Morris 1973; Stein 1956], and asks what it costs when the others' suggestion must be private and computed without a server. The network releases two privatized sums over all of its patients, by secure summation and gossip over the agreement graph. Each hospital turns them, locally and at no further privacy cost, into a leave-one-out prior for its own head and combines it with its own noise-free data, with a weight that its own evidence decides. Privacy noise therefore touches only what a hospital borrows, and is calibrated to the whole network rather than to the smallest hospital. This is how small data can help small data: what a hospital borrows is set by the network's total, not by any one partner, so in twenty hospitals of thirty patients each borrows from the other 570, discounted by how much they differ. The smallest hospitals gain the most, in theory (Section 4.1) and on real networks (Section 7.2).

Contributions. (1) A formal setting: five requirements for a sparse, serverless network of small hospitals, which no existing approach meets together (Sections 2 and 3). (2) Closed-form theory: every hospital gains almost the same number $B = \sigma^2/(\tau^2 + M)$ of virtual patients, privacy enters only through M, and gossip is free post-processing with a rate set by the spectral gap (Section 4). (3) A tuning-free method for likelihood-based heads, GHB, with a privacy guarantee for the whole protocol, and a lighter variant, GHB-lite, that needs no secure summation (Section 5). (4) An evaluation in eighteen simulations against baselines tuned with access to the true loss (Section 6), and on real hospital networks: seven small networks, twelve design cancer networks and eight held-out ones from two consortia, the FLamby benchmark and two image tasks, against six to twelve federated and personalized methods tuned as a consortium could and as no consortium can

(Section 7), with formal tests of the main claims under a plan fixed before they were computed. All code and every number are released.

The paper follows one thread. Section 2 shows what each line of work leaves open, Section 3 turns that into requirements, Section 4 solves the core problem exactly, and Section 5 carries the solution to real model heads. Section 6 tests it where the truth is known, Section 7 where it is not, and Section 8 says when to use it.

2 BACKGROUND AND RELATED WORK

2.1 Small data and siloed smallness

A dataset is small for a task when the information it carries falls short of what the task needs. The companion survey [Simões et al. 2026] measures that information by the *effective sample size*, to first order

$$n^{\text{eff}} = n \cdot \frac{1}{\text{DEFF}} \cdot \rho_L \cdot (1 - 2\eta)^2 \cdot \rho_S, \tag{1}$$

with the design effect DEFF of repeated records [Kish 1965], the labeled fraction ρ_L, the label noise η [Angluin and Laird 1988] and the shift ρ_S to the population of use, and compares it with the sample complexity of the task [Riley et al. 2019]. A task is *siloed-small* when every custodian's data are small for it but their union, after accounting for heterogeneity, is not. The obstacle is then access, not data, and the gain from collaboration is naturally counted in effective patients, which is how Section 4 counts it.

2.2 Distributed statistics and empirical Bayes

Biostatistics fits models across sites without moving records. Random-effects meta-analysis combines site estimates and estimates their heterogeneity τ^2 [DerSimonian and Laird 1986]; distributed regression exchanges sufficient statistics or gradients [Duan et al. 2020; Gaye et al. 2014; Jordan et al. 2019; Wu et al. 2012]; and infrastructures such as OHDSI [Hripcsak et al. 2015] and the Personal Health Train [Beyan et al. 2020; Deist et al. 2020] send analyses to the data. These are exact and auditable, but estimate one population model, offer disclosure controls rather than formal privacy, and mostly assume a coordinator. Empirical Bayes adds the missing step: shrinking each site's estimate toward the population dominates separate estimation [Efron and Morris 1973; Gelman et al. 2013; Stein 1956]. This is GHB's statistical core. What is open is its cost when the population estimate must be private and serverless, and its robustness when τ^2 must be estimated from noisy releases.

2.3 Federated and personalized learning

Federated averaging trains one model by alternating local steps with averaging at a server [McMahan et al. 2017], and has reached near-pooled quality in medical imaging and clinical prediction [Dayan et al. 2021; Pati et al. 2022; Rieke et al. 2020; Sheller et al. 2020]; FLamby provides cross-silo healthcare benchmarks [Ogier du Terrail et al. 2022]. Under heterogeneity the averaged model can be worse than local ones [Hsieh et al. 2020; Zhao et al. 2018], and corrections [Karimireddy et al. 2020; Li et al. 2020, 2021b; Mohri et al. 2019] still produce one model. Personalized methods learn one model per site through multi-task learning, meta-learning, proximal regularization, shared representations or clusters [Collins et al. 2021; Deng et al. 2020; Fallah et al. 2020; Ghosh et al. 2020; Li et al. 2021a; Smith et al. 2017; T. Dinh et al. 2020]; the theory of shared representations explains why a common representation reduces each site to a small head [Du et al. 2021; Maurer et al. 2016; Tripuraneni et al. 2020], the split GHB adopts. They keep the server, and how strongly each site relies on the others is a hyperparameter tuned on validation data that small sites lack. Fairness here has two meanings. Fairness across sites, that no site is badly served [Li et al. 2021a; Mohri et al. 2019], is

what this paper measures throughout, as each hospital's error relative to its own local estimate. Fairness across patient groups [Salazar et al. 2023] needs a protected attribute and a fairness metric; GHB does not target it, and Section 6.6 measures what borrowing does to it.

2.4 Decentralized learning

Decentralized SGD and gossip replace the server by neighbour averaging, with rates set by the spectral gap of the graph [Boyd et al. 2006; Koloskova et al. 2020; Lian et al. 2017; Nedić and Ozdaglar 2009; Xiao and Boyd 2004]; personalized versions use a similarity graph [Bellet et al. 2018; Vanhaesebrouck et al. 2017], and decentralization amplifies privacy [Cyffers and Bellet 2022]. Healthcare examples, Swarm Learning [Warnat-Herresthal et al. 2021], ProxyFL [Kalra et al. 2023] and peer-to-peer segmentation [Roy et al. 2019; Tedeschini et al. 2022], either lack formal privacy by default or share full models, and none models heterogeneity explicitly.

2.5 Privacy in multi-site learning

Gradients and models leak training data [Geiping et al. 2020; Hatamizadeh et al. 2023; Shokri et al. 2017; Zhu et al. 2019]. Differential privacy bounds what any output reveals about one patient [Dwork et al. 2006; Dwork and Roth 2014], extends to deep and federated learning [Abadi et al. 2016; McMahan et al. 2018; Mironov 2017], and composes tightly under zero-concentrated accounting [Bun and Steinke 2016]. Its noise is calibrated to sensitivity, which grows as sample size shrinks, so it costs most for small datasets and large models [Bagdasaryan et al. 2019; Tramèr and Boneh 2021]. Secure aggregation reveals only sums [Bonawitz et al. 2017]. For small hospitals, a method that releases often, releases large objects or calibrates noise to each site cancels the benefit of collaboration.

2.6 What is missing

Each line supplies a piece: exact population estimation, the optimal combination with each site, the representation and head split, serverless computation and a formal guarantee. Missing is their combination: a method that spends privacy once, at the network level, personalizes without tuning, runs over whatever agreements exist, and states its benefit to each hospital in advance.

3 PROBLEM SETTING

There are K hospitals, the vertices of a connected undirected graph $\mathcal{G} = (V, E)$ whose edges are agreements to exchange privatized summaries; $\mathcal{G}$ is sparse ($|E| = O(K)$) and no vertex is trusted by all. Hospital k holds n_k labeled records from its population P_k and wants a model for that population. We require:

(R1) no patient-level record leaves its hospital;
(R2) every released quantity satisfies (ϵ, δ)-differential privacy with respect to the patients of the releasing hospital;
(R3) no central server: computation uses only messages along edges of $\mathcal{G}$;
(R4) the smallest hospitals benefit at least as much as the largest, in relative terms;
(R5) heterogeneity across hospitals is modeled rather than assumed away.

(R4) and (R5) exclude a single global model, which serves the typical site and charges its privacy cost to the smallest. Existing data-exchange agreements do not remove the problem: they mostly serve primary care, and agreements for secondary use form exactly the sparse graph assumed here [European Parliament and Council of the European Union 2016, 2025]. GHB runs over them, and infrastructures such as OHDSI, DataSHIELD and the Personal Health Train [Beyan et al. 2020; Gaye

Table 1. Approaches for sparse, siloed hospital networks against requirements (R1) to (R5), as design properties of each method in its standard form. ✓ satisfied; ○ partially or optionally; ✗ not satisfied. Entries for GHB hold by construction under the assumptions of Sections 4 and 5.4; (R2) and (R3) assume secure summation by hospitals that do not collude (Proposition 5.1).

Approach	(R1) local data	(R2) formal DP	(R3) no server	(R4) small sites gain	(R5) heterogeneity
Centralized pooling	✗	✗	✗	○	○
Federated averaging [McMahan et al. 2017]	✓	○	✗	✗	✗
Personalized FL [Collins et al. 2021; Li et al. 2021a]	✓	○	✗	✓	✓
Swarm Learning [Warnat-Herresthal et al. 2021]	✓	✗	✓	○	✗
ProxyFL [Kalra et al. 2023]	✓	✓	✓	○	○
Synthetic data sharing [Jordon et al. 2019]	✓	○	✓	○	✗
Distributed meta-analysis [DerSimonian and Laird 1986; Duan et al. 2020]	✓	○	○	○	✓
GHB (this paper)	✓	✓	✓	✓	✓

et al. 2014; Hripcsak et al. 2015] can carry its messages. Where a trusted environment for pooling exists, (R3) is moot and the borrowing of Section 4.1 can run inside it without noise.

Threat model. GHB protects patients against honest-but-curious hospitals and any observer of the messages. Its joint releases use secure summation, which assumes that hospitals do not collude to remove their noise shares (Proposition 5.1 states what collusion costs); GHB-lite needs no such assumption. Hospitals follow the protocol; Section 6.4 tests what lies do. Site sizes are public.

Why not anonymize? Anonymization, removing identifiers or generalizing records until each matches k others [Sweeney 2002], is the usual answer to sharing health data, and GHB deliberately does not rely on it. Patient records never leave the hospital, so there is nothing to anonymize; what leaves are masked shares that reveal nothing on their own and two noisy sums over the whole network. And anonymization fails on the data it would apply to: rich health records are re-identified from a handful of attributes [Rocher et al. 2019], and its protection, unlike that of differential privacy, does not survive the combination of several releases or the attacker's background knowledge [Dwork and Roth 2014; Ganta et al. 2008]. Differential privacy instead bounds what any release can reveal about any one patient, whatever else is known, and Section 7.6 measures that bound and the risk that remains on real data.

How existing approaches fare. Table 1 compares design properties, not performance. Pooling violates (R1); federated averaging [McMahan et al. 2017] violates (R3) and, under heterogeneity, (R4) and (R5); personalized learning keeps the server; Swarm Learning [Warnat-Herresthal et al. 2021] lacks formal privacy and ProxyFL [Kalra et al. 2023] shares full models; synthetic data trades utility for privacy [Jordon et al. 2019; Stadler et al. 2022]; meta-analysis ignores representations and graph structure. GHB meets all five by construction, under the assumptions of Propositions 4.1, 4.3 and 5.1; whether it also performs is the subject of Sections 6 and 7.

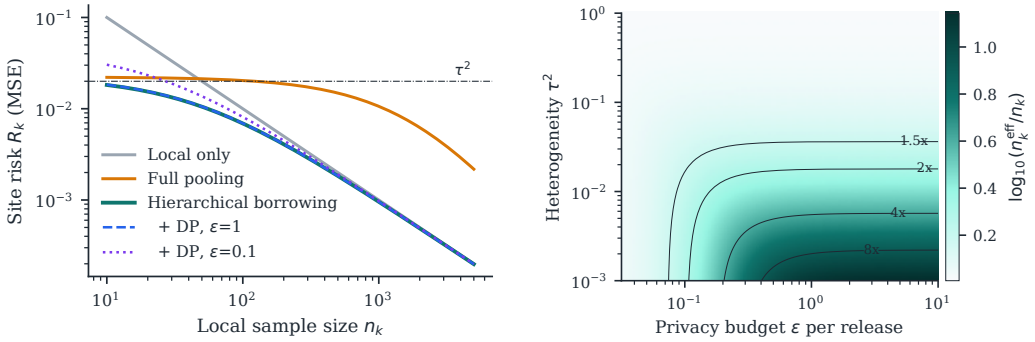


Fig. 1. Left: risk of hospital k as a function of its size n_k, with $K = 12$, other hospitals of 200 patients, $\sigma^2 = 1$, $\tau^2 = 0.02$ and $\delta = 10^{-5}$. Full pooling is excellent for small sites until it hits the heterogeneity floor τ^2, and then harms large ones; hierarchical borrowing (Proposition 4.1), with τ^2 and M known, is never worse than either in expected risk. Privacy at $\epsilon = 1$ is almost free; at $\epsilon = 0.1$ it reduces but does not remove the gain. Right: relative gain in effective sample size n_k^{eff}/n_k for a hospital with $n_k = 50$, as a function of heterogeneity and privacy budget, from Equations (3) and (4).

4 HIERARCHICAL BORROWING IN A MODEL THAT CAN BE SOLVED EXACTLY

We first solve the core problem in the simplest model that keeps its structure: one parameter per hospital, Gaussian random effects, and a population estimate that must be private and serverless. This gives closed forms for what each hospital gains, what privacy costs and what the absence of a server costs.

4.1 Borrowing with an imperfect population estimate

How much should a hospital trust what the others suggest, when their suggestion is itself imperfect? Each hospital has a parameter $\theta_k \in \mathbb{R}$ (one coordinate of a head) and an asymptotically normal local estimate:

$$\theta_k = \mu + u_k, \quad u_k \sim \mathcal{N}(0, \tau^2); \qquad \hat{\theta}_k = \theta_k + e_k, \quad e_k \sim \mathcal{N}(0, v_k), \quad v_k = \sigma^2/n_k, \tag{2}$$

all independent, with heterogeneity τ^2 [DerSimonian and Laird 1986] and σ^2 the inverse Fisher information per patient. Unlike classical empirical Bayes [Efron and Morris 1973; Gelman et al. 2013], hospital k only has an estimate $\tilde{\mu}$ of μ, with error.

Proposition 4.1 (Borrowing with an imperfect prior). *Let $\tilde{\mu}$ be any estimate of μ that is independent of (u_k, e_k) and has mean squared error $M = \mathbb{E}(\tilde{\mu} - \mu)^2$. Among estimators $\tilde{\theta}_k = \alpha\hat{\theta}_k + (1-\alpha)\tilde{\mu}$, the risk $R_k(\alpha) = \mathbb{E}(\tilde{\theta}_k - \theta_k)^2$ is minimized at*

$$\alpha_k^\star = \frac{\tau^2 + M}{\tau^2 + M + v_k}, \qquad R_k^\star = \frac{v_k\,(\tau^2 + M)}{v_k + \tau^2 + M} \;\le\; \min\{v_k,\ \tau^2 + M\}.$$

Equivalently, the effective sample size of hospital k becomes

$$n_k^{\text{eff}} = \frac{\sigma^2}{R_k^\star} = n_k + \underbrace{\frac{\sigma^2}{\tau^2 + M}}_{\text{borrowed patients } B}. \tag{3}$$

Proof. $\tilde{\theta}_k - \theta_k = \alpha e_k + (1-\alpha)(\tilde{\mu} - \mu) - (1-\alpha)u_k$. The three terms are mutually independent, and the first and last have zero mean, so the cross terms vanish; the middle term has second moment

M. Hence $R_k(\alpha) = \alpha^2 v_k + (1-\alpha)^2(\tau^2 + M)$. This is a convex quadratic in α, minimized at $\alpha_k^\star$; substituting gives $R_k^\star$. The bound follows because $R_k^\star$ is half the harmonic mean of v_k and $\tau^2 + M$. Finally $\sigma^2/R_k^\star = \sigma^2/v_k + \sigma^2/(\tau^2 + M) = n_k + B$. □

The proof uses only moments and independence, so normality is not needed. R_k averages over the site effect: a hospital far from the others ($|u_k| \gg \tau$) can lose by borrowing, which is why τ^2 must be learned from the data (Section 4.4). Equation (3) is the central quantity of the method. Every hospital gains almost the same number B of *virtual patients*, whatever its size: a hospital of 30 patients may triple its effective sample, one of 3,000 barely notices, so (R4) holds by construction. These patients come from the network as a whole: M shrinks as the other hospitals' patients add up (Equation (4)), so many small hospitals together can give each of them what one large partner would, and τ^2, how much they differ, sets the ceiling $B \le \sigma^2/\tau^2$. As $\tau^2 \to \infty$, $B \to 0$ and each hospital should act alone; as $\tau^2, M \to 0$, full pooling is optimal. With τ^2 and M known, borrowing is never worse than either (Figure 1).

4.2 The cost of privacy

Proposition 4.1 holds for any $\tilde{\mu}$; privacy (R2) constrains how it is obtained. A hospital's own estimate never leaves it and needs no noise, so privacy affects only M.

Proposition 4.2 (Privatized population estimate). *Suppose $\hat{\theta}_j$ is the mean of per-patient contributions clipped to $[-C, C]$, so that replacing one patient changes $\hat{\theta}_j$ by at most $\Delta_j = 2C/n_j$, and that clipping introduces no bias. Each hospital j releases $\hat{\theta}_j + \xi_j$ with $\xi_j \sim \mathcal{N}(0, s_j^2)$ and $s_j = \Delta_j\sqrt{2\ln(1.25/\delta)}/\epsilon$, which is (ϵ, δ)-differentially private for $\epsilon \in (0,1)$ by the Gaussian mechanism [Dwork and Roth 2014] (for larger ϵ, zero-concentrated calibration, as in Section 5.2, gives noise of the same order). For the leave-one-out average $\tilde{\mu} = \frac{1}{K-1}\sum_{j\neq k}(\hat{\theta}_j + \xi_j)$,*

$$M = \frac{1}{(K-1)^2}\sum_{j\neq k}\left(\tau^2 + \frac{\sigma^2}{n_j} + \frac{8C^2\ln(1.25/\delta)}{n_j^2\,\epsilon^2}\right). \tag{4}$$

Proof. The terms $u_j + e_j + \xi_j$ are independent with zero mean and variances $\tau^2 + \sigma^2/n_j + s_j^2$, and $s_j^2 = 4C^2 \cdot 2\ln(1.25/\delta)/(n_j^2\epsilon^2)$. Independence from (u_k, e_k) holds because k is excluded. □

Privacy noise is divided by $K-1$ and vanishes as the network grows, while the local estimate stays noise-free. But it scales as n_j^{-2}, against n_j^{-1} for sampling error: per-hospital releases are cheap for large hospitals and ruinous for tiny ones. Releasing *sums* removes that cost, because the sensitivity of a sum is set by one patient, not by the smallest hospital; Section 5.2 builds GHB on this. Five conditions delimit the guarantee: (a) neighbouring datasets differ in one patient at one hospital, so a patient held by m hospitals is protected at $(m\epsilon, m\delta)$; (b) site sizes are public; (c) a head in $\mathbb{R}^p$ is released as one vector with ℓ_2 clipping, not coordinate by coordinate; (d) for a head fitted by maximum likelihood, the natural release is the mean score at a public point, with sensitivity $2L/n_j$ for an L-Lipschitz loss ($L = \max\|x\|$ for logistic regression), mapped back to the parameter by public post-processing (Section 5.3); (e) refits compose, and zero-concentrated accounting [Bun and Steinke 2016] keeps that composition tight.

4.3 Computation on a sparse graph without a server

Without a server, hospitals average by gossip [Boyd et al. 2006; Xiao and Boyd 2004]: with a symmetric doubly stochastic W supported on E (for instance Metropolis-Hastings weights $W_{kl} = 1/(1+\max(d_k, d_l))$), each round sets $x^{t+1} = Wx^t$, starting from the privatized releases $x_j^0 = \hat{\theta}_j + \xi_j$.

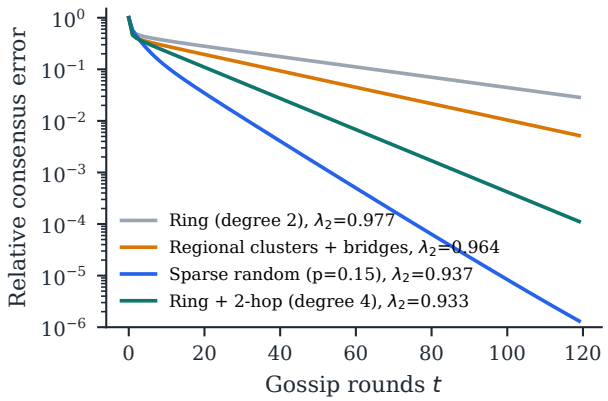


Fig. 2. Gossip consensus error for 24 hospitals on four sparse topologies with Metropolis-Hastings weights. Rates follow the second largest eigenvalue modulus λ of the gossip matrix (Proposition 4.3).

Proposition 4.3 (Free post-processing and convergence). *(i) Every message exchanged during gossip is a function of the privatized releases only, so the whole gossip phase satisfies the same (ϵ, δ) guarantee as the initial release, with no additional privacy cost. (ii) Let $\lambda = \max\{|\lambda_2(W)|, |\lambda_K(W)|\}$. If $\mathcal{G}$ is connected and W has a positive diagonal, then $\lambda < 1$ and*

$$\left\|x^t - \bar{x}\mathbf{1}\right\|_2 \;\le\; \lambda^t \left\|x^0 - \bar{x}\mathbf{1}\right\|_2,$$

so that the estimate held by any hospital after t rounds has root mean squared error at most $\sqrt{M_{\text{all}}} + \lambda^t\sqrt{\mathbb{E}\|x^0 - \bar{x}\mathbf{1}\|_2^2}$ as an estimate of μ, where $M_{\text{all}} = K^{-2}\sum_j(\tau^2 + \sigma^2/n_j + s_j^2)$ is the mean squared error of the full average. (iii) Hospital k obtains its leave-one-out estimate as $\tilde{\mu}_k^t = (Kx_k^t - x_k^0)/(K-1)$. It converges to the leave-one-out average of Proposition 4.2, and differs from it by $K(x_k^t - \bar{x})/(K-1)$, whose root mean square is at most $\frac{K}{K-1}\lambda^t\sqrt{\mathbb{E}\|x^0 - \bar{x}\mathbf{1}\|_2^2}$.

Proof. (i) is the post-processing property of differential privacy [Dwork and Roth 2014]: the gossip iterates are a deterministic function of the released values, and each hospital's subsequent computations on its own data are never released. (ii) Since W is symmetric and doubly stochastic, $\mathbf{1}$ is an eigenvector with eigenvalue 1 and $x^t - \bar{x}\mathbf{1} = W^t(x^0 - \bar{x}\mathbf{1})$ lies in the orthogonal complement of $\mathbf{1}$, on which $\|W\|_2 = \lambda$. Connectivity and a positive diagonal make W irreducible and aperiodic, so $\lambda < 1$ [Xiao and Boyd 2004]. The error bound follows from the triangle inequality in L^2, since each coordinate deviation is bounded by the norm of the whole vector. (iii) Since $\sum_{j\neq k} x_j^0 = K\bar{x} - x_k^0$, the leave-one-out average is $(K\bar{x} - x_k^0)/(K-1)$, and subtracting it from $\tilde{\mu}_k^t$ leaves $K(x_k^t - \bar{x})/(K-1)$, which is bounded as in (ii). □

The rounds needed for accuracy γ are about $\log(1/\gamma)/(1-\lambda)$, so the spectral gap sets the communication cost, and a hospital should use its estimate only once λ^t is small. Which agreements a network signs matters: links that skip one neighbour triple the gap of a ring, and regional clusters joined by single bridges mix little better than a ring (Figure 2).

4.4 Unknown heterogeneity

The propositions take τ^2 as known. How a hospital handles not knowing it decides whether borrowing helps or harms it.

Remark 4.4 (Estimating τ^2, or integrating it out). Moment estimators of τ^2 [DerSimonian and Laird 1986] use only the public releases and their variances, so they inherit the guarantee. Better, hospital k can integrate τ out: take Q equally weighted quantiles τ_q of its posterior under a half-Cauchy prior, and average its estimates under the priors $\mathcal{N}(\tilde{\mu}_k(\tau_q), \tau_q^2 + M_k(\tau_q))$, each weighted by the marginal likelihood of its own data (closed form in the Gaussian model, Laplace otherwise). A hospital whose data agree with the network borrows strongly; one whose data conflict puts its weight on wide components and falls back to its own data. This costs no privacy, since the hospital's data are used only inside it.

With noisy releases the posterior of τ is wide: a plug-in posterior mean then overstates heterogeneity and switches borrowing off, and a plug-in median or moment estimate can understate it. Averaging over the posterior, weighted by each hospital's evidence, commits to neither, and Simulation 16 finds it among the largest improvements (Section 6.5).

4.5 Structured heterogeneity

When similarity has structure, for instance by region, a Gaussian Markov random field prior [Rue and Held 2005] on a similarity graph with weights a_{kl} and Laplacian L gives the maximum a posteriori problem

$$\min_{\theta} \sum_k \frac{(\theta_k - \hat{\theta}_k)^2}{2v_k} + \frac{\beta}{2} \sum_{k<l} a_{kl}(\theta_k - \theta_l)^2 + \frac{\gamma}{2} \sum_k (\theta_k - \mu_0)^2, \tag{5}$$

solved by the Jacobi iteration $\theta_k \leftarrow (\hat{\theta}_k/v_k + \beta \sum_l a_{kl}\theta_l + \gamma\mu_0)/(1/v_k + \beta d_k + \gamma)$, which uses only neighbour values and converges because the system is strictly diagonally dominant. It generalizes Vanhaesebrouck et al. [2017] with an anchor μ_0 from published knowledge; neighbours exchange values computed from privatized releases, and each hospital applies the last update with its own data locally.

5 GHB FOR MODEL HEADS

The analysis applies to any parametric head, in particular the last layer on top of a fixed representation. This section places borrowing in an architecture, specifies what the network releases and how each hospital uses it, and states the privacy of the whole protocol.

5.1 Architecture

GHB keeps the privatized part of the model small and surrounds it with components that cost no privacy (Figure 3), each drawing on a different source of the information a small hospital lacks [Simões et al. 2026]:

- **L0 Public representation:** a foundation encoder pretrained on public data [Moor et al. 2023; Rasmy et al. 2021; Zhou et al. 2023].
- **L1 Local adaptation:** low-rank adapters [Hu et al. 2022] trained self-supervised on the hospital's unlabeled data [Chen et al. 2020; He et al. 2022].
- **L2 Hierarchical head:** a low-dimensional head $\theta_k \in \mathbb{R}^p$ that borrows by Sections 4 and 5.2.
- **L3 Knowledge anchor:** a prior mean μ_0 from published models, so that a hospital with no neighbours starts somewhere sensible.
- **L4 Site calibration:** conformal prediction on local held-out data [Angelopoulos and Bates 2023; Lu et al. 2023; Shafer and Vovk 2008].
- **L5 Targeted procurement:** posterior variances point to the patients most worth labeling next [Settles 2009].

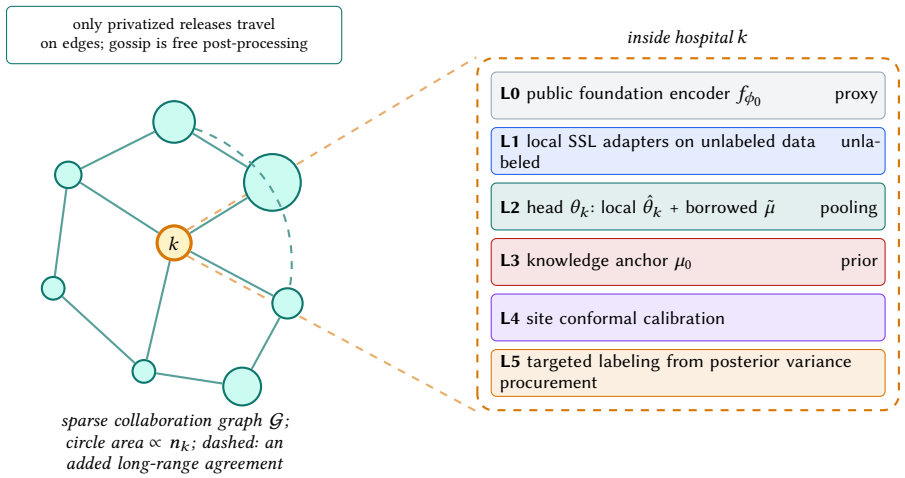


Fig. 3. Graph Hierarchical Borrowing (GHB). Left: hospitals of very different sizes connected by a sparse set of agreements, without a central server. Right: the model held by each hospital, with the source of information used at each layer. Only the low-dimensional head participates in privatized exchange.

A small private component is a necessity, not a convenience: Gaussian noise grows with $\sqrt{p}$, and private learning is far more accurate on good fixed features than end to end [Tramèr and Boneh 2021]. Synthetic data is left out of the training path, since a generator fitted to a hospital's data adds no information about its target [Simões et al. 2026; Stadler et al. 2022].

5.2 What the network releases

Section 4.2 showed that noise calibrated to a sum is set by one patient, not by the smallest hospital. GHB applies this to the whole network: it makes two joint releases, sums over every patient of every hospital, revealed only as totals, and computes everything else from them, from public information and, inside each hospital, from its own data.

Two joint releases. Let $G = \max \|x\|$ bound the features of a likelihood-based head. For logistic regression the per-patient score at the origin is $\psi_i = x_i(y_i - \frac{1}{2})$ and the information is $\phi_i = x_i x_i^\top/4$. The network releases

$$\tilde{S} = \sum_k \sum_{i \in k} \psi_i + \xi, \quad \xi \sim \mathcal{N}(0, s_S^2 I_p), \qquad \tilde{J} = \sum_k \sum_{i \in k} \phi_i + E, \quad E_{ab} = E_{ba} \sim \mathcal{N}(0, s_J^2) \ (a \le b).$$

Replacing one patient moves $\sum \psi_i$ by at most $2G$, a bound valid at any evaluation point (at the origin, where $|y - \frac{1}{2}| = \frac{1}{2}$, it is conservative by a factor of two). Since the ϕ_i are positive semidefinite with Frobenius norm at most $G^2/4$, and $\|A - B\|_F^2 \le \|A\|_F^2 + \|B\|_F^2$ for positive semidefinite A, B, it moves the upper triangle of $\sum \phi_i$ by at most $\sqrt{2}\, G^2/4$. Each release gets half of a zero-concentrated budget ρ, so $s_S = 2G/\sqrt{\rho}$ and $s_J = \sqrt{2}\, G^2/(4\sqrt{\rho})$, with $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$ [Bun and Steinke 2016]. When the matrix would be mostly noise, only its diagonal is released, by a public rule fixed in advance: the diagonal when $s_J \sqrt{p} > NG^2/(4p)$. The sums are computed by secure summation [Bonawitz et al. 2017]: every pair of hospitals derives a pairwise mask from a key exchange relayed over $\mathcal{G}$, the masks cancel in the total, every hospital adds its masked contribution and a share $\mathcal{N}(0, s^2/K)$ of the noise, and the masked shares are added exactly along a spanning tree of $\mathcal{G}$ (in a

finite group), with the total sent back down the tree. No server is needed, and a hospital's share, masked by its masks with all other hospitals, reveals nothing on its own.

From the releases to the network. With $A = (\tilde{J}^+ + I/s_0^2)^{-1}$, where $\tilde{J}^+$ is $\tilde{J}$ with its eigenvalues floored at $0.01N$ (positive definite) and $s_0 = 2.5$ is the weakly informative default scale for logistic coefficients [Gelman et al. 2008], the network estimate is one Newton step of the pooled model from the origin, $\hat{w} = A\tilde{S}$, with public privacy covariance $V_{\mathrm{dp}} = s_S^2 AA^\top$. The privacy part of M is now of order $1/N^2$ per unit of information, with $N = \sum_k n_k$: the smallest hospital no longer sets the noise. Two optional sources of context enter by post-processing: when several networks fit the same head (one per cancer type, say), each $\hat{w}$ is shrunk toward the precision-weighted mean of the others', with a half-Cauchy between-network scale; and a published model μ_0 with stated uncertainty is combined with $\hat{w}$ unless the two conflict ($|\mu_0 - \hat{w}| > 3\sqrt{\sigma_0^2 + V}$ in some coordinate).

From the network to each hospital. Hospitals share how outcomes depend on features, but their base rates differ with case mix and referral, and borrowing base rates biases exactly the small, atypical hospitals. GHB therefore does not borrow the intercept: in every borrowing component it gets an extra prior variance s_0^2, and its default scale stays s_0. Hospital k computes its exact score $s_k = \sum_{i\in k} x_i(y_i - \sigma(x_i^\top \hat{w}))$ and its information at the origin, $J_k = \sum_{i\in k} x_i x_i^\top/4$, consistent with $\tilde{J}$, and removes its own data by one Newton step,

$$\mu_{-k} = \hat{w} - H_{-k}^{-1} s_k, \qquad V_{-k} = H_{-k}^{-1} + V_{\mathrm{dp}}, \qquad H_{-k} = (\tilde{J} - J_k)^+ + I/s_0^2,$$

the leave-one-out estimate of Proposition 4.1 with its full covariance, sampling error and privacy noise. It then integrates the heterogeneity out (Remark 4.4): for $Q = 12$ quantiles τ_q of a half-Cauchy prior with scale 0.5 and default scales $s \in \{0.5, 1, 2.5\}$, it forms the priors $\mathcal{N}(\mu_{-k}, \mathrm{diag}(\tau_q^2) + s_0^2 e_1 e_1^\top + V_{-k}) \cdot \mathcal{N}(0, D_s)$, with e_1 the intercept and $D_s = \mathrm{diag}(s_0^2, s^2, \ldots, s^2)$, adds three non-borrowing components $\mathcal{N}(0, D_s)$ with a quarter of the weight, fits its head under each, and averages the fits by the Laplace approximation of its own marginal likelihood. Agreement with the network puts the weight on narrow components; conflict puts it on wide or non-borrowing ones. Every setting is fixed in advance; none is tuned on outcome data.

5.3 GHB-lite: one release per hospital

Where secure summation is unavailable, GHB-lite has each hospital release its own mean score at the anchor,

$$g_k = \frac{1}{n_k}\sum_{i\in k} x_i\big(y_i - \sigma(x_i^\top \mu_0)\big) + \xi_k, \qquad \xi_k \sim \mathcal{N}(0, s_k^2 I_p), \quad s_k = \frac{2G}{n_k\sqrt{2\rho}},$$

private on its own whatever the others do. Under a public feature distribution, $\mathbb{E}g_k = m(\theta_k) - m(\mu_0)$ with $m(\theta) = \mathbb{E}[x\,\sigma(x^\top\theta)]$, so the release is mapped to the parameter scale by moment matching, $\tilde{\theta}_k = a_k + I(a_k)^{-1}(g_k + m(\mu_0) - m(a_k))$, linearized at the pooled and then the leave-one-out estimates. Each hospital forms the posterior of τ from the mapped releases, gossips their inverse-variance-weighted average for each quantile, and combines it with the anchor and its own evidence exactly as above. Evaluating the score at the anchor makes it a control variate: a wrong anchor makes the release noisier, not biased.

The two protocols trade assumptions. GHB-lite needs only honest-but-curious neighbours and learns τ from the spread of the releases, which GHB cannot, since it releases nothing per hospital. But its moment matching is exact only when every hospital's features follow the public distribution, and its noise, calibrated to $1/n_k$, swamps small hospitals. GHB uses the hospitals' real information

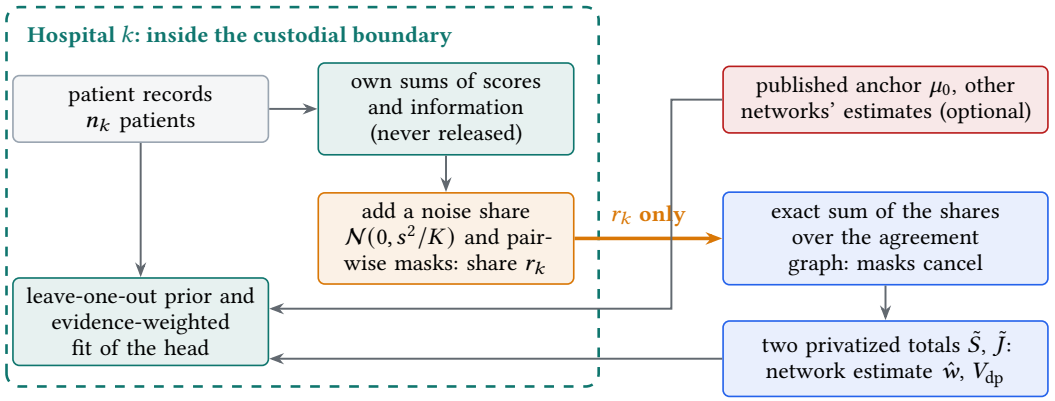


Fig. 4. The privacy boundary of GHB. Patient records and the local head stay inside the hospital. Only the hospital's masked share r_k of the two network sums crosses the boundary; the masks cancel in the totals, which carry the privacy noise. Everything computed from the totals is post-processing (Proposition 5.1), and the hospital combines it with its own noise-free data locally.

Algorithm 1: Graph Hierarchical Borrowing (one round of model fitting)

Input: graph $\mathcal{G}$ with gossip matrix W; privacy budget (ϵ, δ), i.e. ρ; feature bound G; public encoder f_{ϕ_0}; optionally a published model μ_0 with stated uncertainty σ_0^2, and the estimates of other networks fitting the same head

foreach *hospital* k *in parallel* **do**
  adapt f_{ϕ_0} to local unlabeled data with adapters (L1);
  compute $\sum_{i\in k} x_i(y_i - \frac{1}{2})$ and $\sum_{i\in k} x_i x_i^\top/4$ (its diagonal alone if $s_J\sqrt{p} > NG^2/(4p)$); add pairwise masks agreed with every other hospital and a share $\mathcal{N}(0, s^2/K)$ of the noise, $s_S = 2G/\sqrt{\rho}$, $s_J = \sqrt{2}\,G^2/(4\sqrt{\rho})$; `// the only released quantities`
add the masked shares along a spanning tree of $\mathcal{G}$ and send the total back: every hospital obtains $\tilde{S}$ and $\tilde{J}$; `// the masks cancel`
compute $\hat{w} = A\tilde{S}$ and $V_{\text{dp}} = s_S^2 AA^\top$, $A = (\tilde{J}^+ + I/s_0^2)^{-1}$; shrink $\hat{w}$ toward other networks' estimates, and combine it with μ_0 unless they conflict; `// post-processing`
foreach *hospital* k **do**
  compute locally s_k at $\hat{w}$ and J_k; leave-one-out estimate $\mu_{-k} = \hat{w} - H_{-k}^{-1}s_k$, $V_{-k} = H_{-k}^{-1} + V_{\text{dp}}$;
  fit the local head under each prior $\mathcal{N}(\mu_{-k}, \text{diag}(\tau_q^2) + s_0^2 e_1 e_1^\top + V_{-k}) \cdot \mathcal{N}(0, D_s)$, $q = 1, \ldots, Q$, $s \in \{0.5, 1, 2.5\}$, and under $\mathcal{N}(0, D_s)$ alone (the intercept is not borrowed); average the fits weighted by the marginal likelihood of the local data (Remark 4.4); in the Gaussian case with known τ^2 this is $\tilde{\theta}_k = \alpha_k^\star \hat{\theta}_k + (1 - \alpha_k^\star)\tilde{\mu}_k$ with $\alpha_k^\star$ of Proposition 4.1;
  calibrate with conformal prediction on local held-out data (L4); rank unlabeled patients by posterior variance (L5);

and network-level noise, at the price of trusting the summation. Which matters more is an empirical question, and the answer differs between simulation and real hospitals (Sections 6 and 7).

5.4 The protocol and its privacy boundary

Algorithm 1 assembles the protocol, and Figure 4 shows what crosses the custodial boundary: masked shares of two network sums, whose totals are privatized, and nothing else.

Proposition 5.1 (Privacy of GHB). *Suppose every patient's features satisfy $\|x\| \le G$, the secure summation reveals only the two totals, and the noise of each total has the stated variance. Then the pair $(\tilde{S}, \tilde{J})$ satisfies ρ-zero-concentrated differential privacy, and hence (ϵ, δ)-differential privacy with $\epsilon = \rho + 2\sqrt{\rho \ln(1/\delta)}$, with respect to replacing the record of any one patient at any hospital, against any observer outside the network. Every model, estimate and message that GHB computes afterwards satisfies the same guarantee. A participating hospital knows its own noise share and can remove it, so against a single hospital the guarantee for the other hospitals' patients is $\rho K/(K-1)$, and against c colluding hospitals $\rho K/(K-c)$.*

Proof. Replacing one record changes $\sum \psi_i$ by at most $2G$ and the upper triangle of $\sum \phi_i$, or its diagonal, by at most $\sqrt{2}\,G^2/4$ in ℓ_2 norm, as shown above; whether the diagonal or the full matrix is released depends only on public quantities. The Gaussian mechanism with noise s and sensitivity Δ is $\Delta^2/(2s^2)$-zCDP [Bun and Steinke 2016], which is $\rho/2$ for each total; zCDP composes additively, so the pair is ρ-zCDP, and the conversion to (ϵ, δ) is that of Bun and Steinke [2016]. Everything computed afterwards is a function of the two totals, of public quantities, and, at hospital k, of hospital k's own data used only inside it and never released, so post-processing [Dwork and Roth 2014] gives the second claim; the networks' exchange of their estimates is post-processing as well, and each masked share, uniformly distributed given the data as long as one other hospital keeps its pairwise mask secret, reveals nothing beyond the totals. A hospital, or c colluding hospitals, who subtract their own shares leave noise of variance $s^2(K-c)/K$, for which the same computation gives $\rho K/(K-c)$; for $c = 1$ and the networks studied here ($K \ge 5$) that is at most 1.25ρ. A network that wants ρ against its own members can scale the noise by $\sqrt{K/(K-1)}$. □

A hospital's final model uses its own data without noise and is therefore not private; like the local estimate in Proposition 4.1, it never leaves the hospital. GHB-lite satisfies the same guarantee per hospital with no trust in summation. None of GHB's ingredients is new: shrinkage dominates separate estimation [Efron and Morris 1973; Stein 1956], noise on sums is the cheapest privacy [Bonawitz et al. 2017; Dwork and Roth 2014], gossip removes the server [Boyd et al. 2006], evidence weighting lets borrowing switch itself off, and shared representations keep the head small [Maurer et al. 2016; Tripuraneni et al. 2020]. Its case is that it is the simplest combination that meets all five requirements; the next two sections test whether it also performs.

6 EXPERIMENTS

The propositions assume τ^2 and M known, a Gaussian model and honest participants. Eighteen Monte Carlo simulations, where the truth is known, measure what is lost when each assumption fails; Section 7 then asks the same of real hospitals. Figure 5 places every scenario on one axis and Table 2 lists questions and answers; the supplement gives each design in full.

6.1 Design of the simulation study

Unless stated otherwise the network has 24 hospitals of 16 to 479 patients (log-normal, median 100), $\sigma^2 = 1$, $C = 1$, $\epsilon = 1$, $\delta = 10^{-5}$, and releases are averaged by 60 gossip rounds on the ring with two-hop links of Figure 2. Errors are relative to local estimation, so values below 1 mean that hospitals gain. The simulations were built for GHB-lite, whose theory is that of Section 4; GHB was added to the logistic comparisons (Simulations 8, 10, 11 and 15) without changing anything else, and has its own ablation and sensitivity analyses (Simulations 16B and 17B). These worlds favour GHB-lite on purpose: its public feature distribution is exactly that of every hospital, and 24 hospitals of about a hundred patients make its releases informative enough to learn τ. Every baseline spends the same budget under zero-concentrated accounting, and its hyperparameters

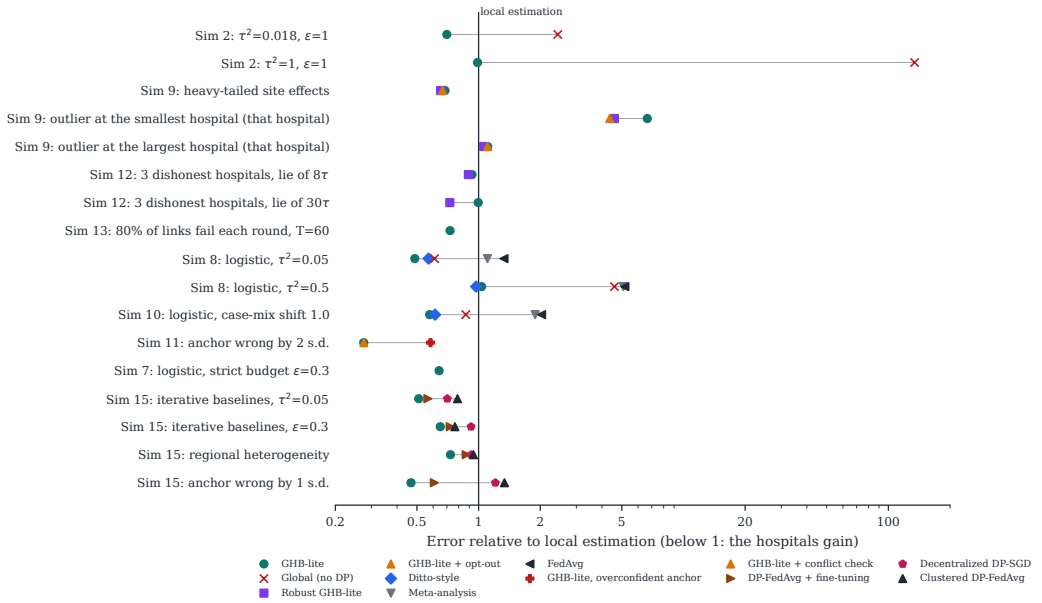


Fig. 5. Summary of Simulations 2 to 15. Each row is a scenario and each marker a method; the horizontal position is the method's error relative to local estimation (mean risk R_k/v_k in the Gaussian simulations, mean excess log-loss divided by that of local estimation in the logistic ones). Left of the vertical line, the hospitals gain. Rows marked "that hospital" report the outlying hospital alone.

are tuned on separate replicates with access to the true loss, which no deployment has; GHB and GHB-lite tune nothing. Results are means with ±2 standard errors.

6.2 Does the theory survive estimation?

A real network must estimate τ^2 and M from privatized releases, so the first question is how much of the predicted gain survives, and for whom. Simulation 1 confirms the closed forms. Across four orders of magnitude of heterogeneity (Simulation 2, Figure 6), plug-in GHB-lite is never worse than local estimation for any hospital by more than Monte Carlo error (worst 1.02), while a global model becomes up to 134 times worse on average and 382 times worse for its worst hospital; the price of estimating τ^2 appears only when heterogeneity is tiny (0.12 against 0.07 with the true value). Gossip is free once mixed: 11 to 35 rounds suffice on the topologies of Figure 2 (Simulation 4), and mixing survives 80% of links failing each round (0.73 against 0.72, Simulation 13).

Who gains is what (R4) asks. Every hospital gains about $B \approx 46$ virtual patients (Simulation 3, Figure 7), so hospitals of 60 patients or fewer multiply their effective sample size by 2.28 and those of 300 or more by 1.08; a global model helps small hospitals less (1.44) and cuts large ones to 0.15 of their own. With three hospitals, τ^2 is too poorly estimated and most of the gain is lost; a half-Cauchy prior helps from four upward.

6.3 Against federated and meta-analytic baselines

Gaining over local estimation matters only if the method also beats what a network could adopt instead. From here on the heads are logistic regressions on four features, every baseline consumes the same budget, and its hyperparameters are tuned on the true loss.

Table 2. The eighteen simulations, the held-out real networks, and their main results. The simulations run GHB-lite; where GHB was added, its value follows ("GHB"). Relative risk is R_k/v_k and logistic results are mean excess log-loss (nats); below the local value, hospitals gain. $\epsilon = 1$ unless stated.

Simulation	Question	Result
1 Closed forms	Do Propositions 4.1 and 4.2 match simulation?	98% of 101 comparisons within two standard errors; largest relative deviation 1.6%
2 Heterogeneity	Is plug-in GHB-lite ever worse than local estimation?	Worst hospital at most 1.02 for $\tau^2 \in [10^{-4}, 1]$; a global model reaches 382
3 Equity, network size	Who gains, and how many hospitals are needed?	Gain in n^{eff} 2.28 for small, 1.08 for large hospitals; with 3 hospitals relative risk 0.91 against 0.78 with known τ^2
4 Gossip rounds	How many rounds are needed?	Within 2% of exact averaging after 11 to 35 rounds, against 69 to 203 from the worst-case bound
5 Joint release	Does secure aggregation among small hospitals help?	Only if all hospitals are small: B from 1.0 to 6.9 at $\epsilon = 0.1$; from 19.3 to 21.0 in a mixed network
6 Regional structure	When does the GMRF prior of Equation (5) pay off?	Worse when regions barely differ (0.40 against 0.31), far better when they differ (0.40 against 0.94)
7 Logistic heads	Does it work beyond the Gaussian model?	0.0266 local, 0.0135 GHB-lite, 0.0173 pooled without privacy; 0.0175 against 0.0273 at $\epsilon = 0.3$
8 Federated baselines	Does it beat existing approaches at the same budget?	At $\tau^2 = 0.05$: GHB-lite 0.0134, GHB 0.0184, Ditto-style [Li et al. 2021a] 0.0156, meta-analysis [DerSimonian and Laird 1986] 0.0303, FedAvg [McMahan et al. 2017] 0.0365; at $\tau^2 = 0.5$ GHB-lite 0.0282 against local 0.0273
9 Atypical hospitals	What happens to a hospital unlike the others?	Heavy tails: 0.69, robust prior 0.65; an outlier at the smallest hospital loses (6.7, at best 4.4), at the largest barely (1.11)
10 Case-mix shift	Does it survive hospitals with different patients?	Shift of one s.d.: GHB-lite 0.0132, GHB 0.0164, Ditto-style [Li et al. 2021a] 0.0140, pooled 0.0198, local 0.0228
11 Wrong anchor	What if the published model is wrong?	Honest uncertainty or a conflict check: at most 0.0167 at any error; an over-confident anchor 0.0352 at 2 s.d.; GHB at most 0.0198
12 Dishonest hospitals	What if hospitals lie?	Plain GHB-lite never above 1.00; robust screening keeps 0.72 against large lies, at a cost of 0.81 against 0.73 when nobody lies
13 Link failures	What if links fail during gossip?	With 80% of links failing each round, 0.73 after 60 rounds against 0.72 with exact averaging
14 Repeated releases	What do refits on the same patients cost?	After 64 refits, $B = 1.5$ with basic composition and 32.4 with zero-concentrated accounting (one release: 46.1)
15 Iterative federated learning	Does it beat multi-round training with the same budget and tuned hyperparameters?	At $\tau^2 = 0.05$: GHB-lite 0.0138, GHB 0.0192, DP-FedAvg [McMahan et al. 2018] with fine-tuning 0.0153, decentralized DP-SGD [Lian et al. 2017] 0.0191, clustered [Ghosh et al. 2020] 0.0214; at $\epsilon = 0.3$ GHB-lite 0.0175 against 0.0196; worst hospital at most 1.06 for GHB-lite except at $\tau^2 = 0.5$, 1.39 to 2.96 for fine-tuning
16 Ablation	What does each component contribute?	Error relative to GHB-lite without it: ridge release up to 1.39; $X^\top y$ release 1.74 under case-mix shift; plug-in τ^2 up to 1.37 (posterior mean) and 1.33 (DerSimonian-Laird [DerSimonian and Laird 1986]); no conflict check 1.36 with a wrong anchor; equal weights up to 1.86; gossip and accounting free
17 Sensitivity	How much do settings and conditions matter?	Settings: quantiles, public sample, threshold and rounds ($\geq$ 20) flat; prior scale of τ 0.90 to 1.17; misstated anchor up to 1.41. Conditions: error 0.40 to 0.69 of local, worst hospital at most 1.03
16B, 17B GHB	Which of GHB's components and settings matter, and across which conditions?	Intercept borrowed 0.72 to 1.01 of GHB's error (this world shares base rates); no non-borrowing components down to 0.93; covariance, leave-one-out, sensitivity and $Q \geq 3$ within 1%, heavier non-borrowing weight up to 1.24; across conditions 0.64 to 0.82 of local, worst hospital at most 1.11
18 Group fairness	What does borrowing do to a protected group?	Minority excess log-loss ($\times 10^{-3}$) with a shared disparity: GHB-lite 17.8, local 34.3, FAIR-FATE [Salazar et al. 2023] 30.3; disparities that vary: GHB-lite 27.3, global 60.7; equalized odds of GHB-lite within 0.02 of the true models, FAIR-FATE closer to parity
Real data	Does it hold on real hospitals, against the literature tuned as a consortium could?	Eight held-out networks (TCGA and CPTAC): GHB 0.916 and 0.907 relative to local at $\epsilon = 1$ and 10; strongest realistic literature method 0.950 and 0.911 (Section 7)

One-shot baselines. Simulations 8 and 10 compare methods that use the same releases (Figure 8, Table 3). At $\tau^2 = 0.05$ GHB-lite reaches 0.0134, below a Ditto-style [Li et al. 2021a] model tuned on separate replicates (0.0156) and below a global model trained on pooled data without privacy (0.0167); federated averaging [McMahan et al. 2017] and meta-analysis [DerSimonian and Laird 1986], which give every hospital one model, are worse than local (0.0365 and 0.0303 against 0.0274). Under case-mix shift GHB-lite keeps its lead (0.0132 against 0.0140). From $\tau^2 = 0.2$ the tuned

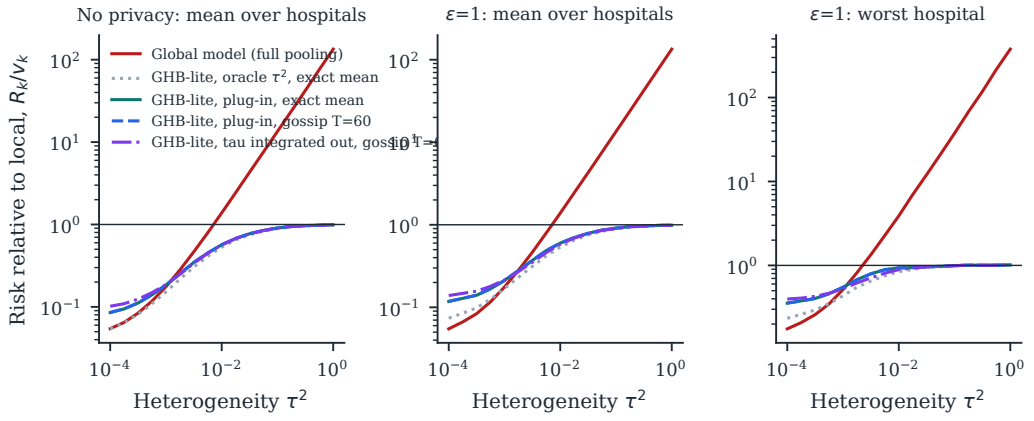


Fig. 6. Simulation 2: risk relative to local estimation against heterogeneity, for 24 hospitals of 16 to 479 patients. Plug-in GHB-lite estimates τ^2 and M from privatized releases averaged by gossip; the oracle uses the true τ^2; the global model is the sample-size weighted mean of all site estimates, computed without privacy.

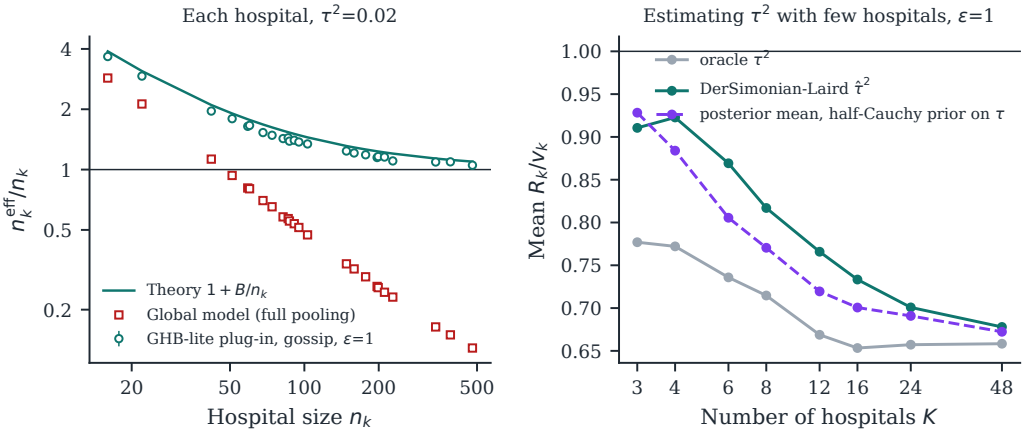


Fig. 7. Simulation 3. Left: effective sample size relative to own size for each hospital at $\tau^2 = 0.02$, $\epsilon = 1$; the line is $1 + B/n_k$ from Equation (3). Right: mean relative risk against the number of hospitals for three ways of obtaining τ^2.

Ditto-style model is slightly better, and at $\tau^2 = 0.5$ it is the only method that gains. GHB is less accurate than GHB-lite here (0.0184 at $\tau^2 = 0.05$), for two reasons that are this world's: it cannot learn τ from per-hospital releases, and these hospitals differ in base rate no more than in slopes, so not borrowing the intercept costs rather than protects.

Iterative federated learning. Simulation 15 faces multi-round training: DP-FedAvg [McMahan et al. 2017, 2018] with and without fine-tuning, decentralized DP-SGD [Lian et al. 2017] on the same graph, and clustered training in the style of IFCA [Ghosh et al. 2020], all tuned on 20 replicates with the true loss (Figure 9, Tables 4 and 5); four of the ten settings were not used while GHB-lite was revised. GHB-lite is the most accurate private method in six settings, three of them held out,

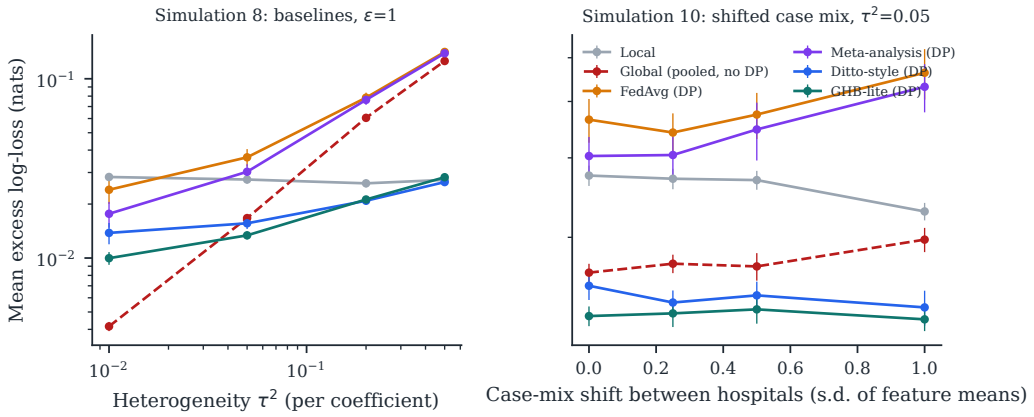


Fig. 8. Simulations 8 and 10, logistic-regression heads. Mean excess log-loss over the true model of GHB-lite, of private baselines that use the same releases (one-shot federated averaging [McMahan et al. 2017], random-effects meta-analysis [DerSimonian and Laird 1986], Ditto-style [Li et al. 2021a] personalization tuned on separate replicates) and of a global model trained on the pooled data without privacy. Left: against coefficient heterogeneity. Right: against case-mix shift between hospitals at $\tau^2 = 0.05$. Bars: ±2 standard errors.

Table 3. Simulations 8 and 10: exact values behind Figure 8. Values are mean excess log-loss in 10^{-3} nats, with two Monte Carlo standard errors in grey; the lowest private method in each row is bold. All private methods use the same releases at $\epsilon = 1$; the pooled model uses no privacy. GHB (joint releases) appears in the table but not in the figure.

Setting	Local	GHB	GHB-lite	Ditto-style [Li et al. 2021a]	Meta-analysis [DerSimonian and Laird 1986]	FedAvg [McMahan et al. 2017]	Pooled, no DP
$\tau^2 = 0.01$	28.3 (1.1)	16.4 (1.3)	**10.0** (0.8)	13.8 (1.9)	17.7 (2.9)	24.0 (4.0)	4.2 (0.2)
$\tau^2 = 0.05$	27.4 (1.4)	18.4 (1.1)	**13.4** (0.7)	15.6 (1.1)	30.3 (3.1)	36.5 (4.1)	16.7 (0.8)
$\tau^2 = 0.2$	26.1 (1.3)	24.0 (1.3)	21.2 (0.9)	**20.9** (0.9)	76.3 (4.7)	78.4 (5.2)	60.5 (2.3)
$\tau^2 = 0.5$	27.3 (1.4)	30.2 (1.6)	28.2 (1.3)	**26.5** (1.1)	138.8 (5.1)	141.1 (5.2)	125.6 (4.3)
shift 0.25	27.0 (1.4)	18.6 (1.1)	**13.6** (0.9)	14.3 (0.9)	30.4 (3.0)	34.1 (3.5)	17.5 (0.8)
shift 0.5	26.8 (1.3)	18.4 (1.1)	**13.8** (1.0)	14.9 (1.1)	34.7 (5.1)	37.4 (4.4)	17.2 (1.2)
shift 1.0	22.8 (1.0)	16.4 (1.1)	**13.2** (0.8)	14.0 (1.2)	43.1 (5.3)	46.3 (5.9)	19.8 (1.2)

for instance at $\tau^2 = 0.05$ (0.0138 against 0.0153 for DP-FedAvg with fine-tuning) and at $\epsilon = 0.3$ (0.0175 against 0.0196); the tuned Ditto-style model is slightly better at $\tau^2 = 0.2$, without privacy and with a wrong anchor, and at $\tau^2 = 0.5$ nothing beats local estimation. Its gain at a strict budget comes from borrowing, not the anchor: a tuned local control started from the anchor reaches 0.0205 and becomes worse than local when the anchor is wrong (0.0419). GHB, with joint releases, does less well in this world, whose hospitals share their base rates: significantly better than local estimation in 9 of 10 settings, but significantly worse than the tuned Ditto-style model in 8 and than DP-FedAvg with fine-tuning in 4 (supplement).

The methods differ most in who pays. GHB-lite's worst hospital stays within a factor 1.06 of its local error in every private setting except $\tau^2 = 0.5$; fine-tuning leaves some hospital 1.39 to 2.96 times worse off, usually a large one whose data the federated model overrides (the eight largest go from 0.0094 alone to 0.0131 at $\epsilon = 0.3$, against 0.0079 with GHB-lite). What remains is the cost of not knowing τ^2: with the true value GHB-lite would reach 0.0158 at $\epsilon = 0.3$. GHB, for the reasons above, beats every iterative baseline in 3 of the ten settings (one by a near tie) and trails GHB-lite

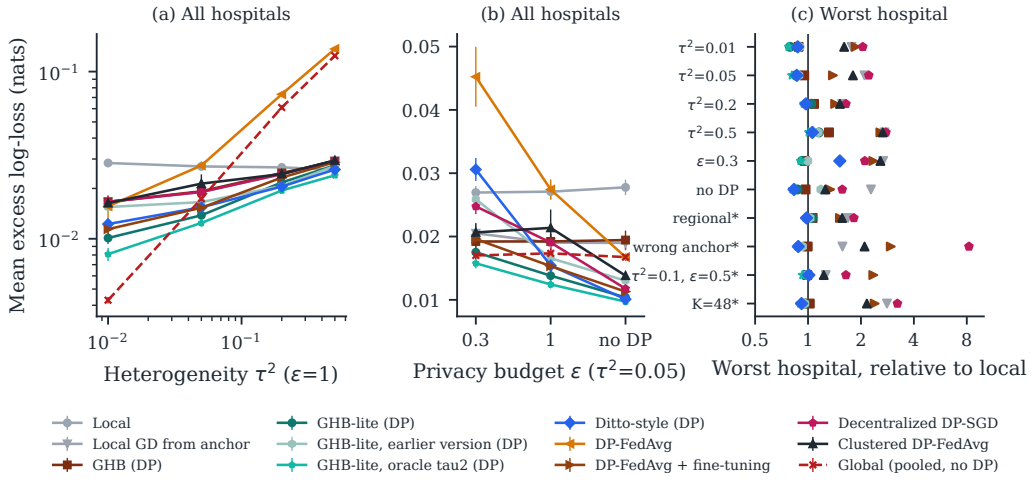


Fig. 9. Simulation 15: GHB-lite and GHB against iterative federated learning at the same privacy budget (logistic heads, 24 hospitals, mean excess log-loss over 60 replicates, ±2 standard errors). Baseline hyperparameters are tuned with access to the true loss; GHB-lite is not tuned. (a) Against heterogeneity at $\epsilon = 1$. (b) Against the privacy budget at $\tau^2 = 0.05$. (c) The worst hospital's error relative to its own local estimate in each setting; right of 1, some hospital is worse off than alone; settings marked $*$ were not used while GHB-lite was revised. "GHB-lite, earlier version" is an earlier formulation that released ridge fits, shown for reference; "GHB-lite, oracle τ^2" uses the true heterogeneity and is a reference, not a method.

Table 4. Simulation 15: exact values behind Figure 9. Values are mean excess log-loss in 10^{-3} nats, with two Monte Carlo standard errors in grey; the lowest private method in each row is bold. Settings marked $*$ were not used while GHB-lite was revised. FT: fine-tuning [McMahan et al. 2018]. In the figure, "GHB-lite (DP)" is GHB-lite and "GHB (DP)" is GHB with joint releases.

Setting	Local	Anchor GD	GHB	GHB-lite	Ditto-style [Li et al. 2021a]	DP-FedAvg + FT [McMahan et al. 2018]	Decentr. DP-SGD [Lian et al. 2017]	Clustered DP-FedAvg [Ghosh et al. 2020]	GHB-lite, true τ^2	Pooled, no DP
$\tau^2 = 0.01$	28.4 (1.3)	16.6 (1.5)	16.6 (1.1)	**10.1** (0.8)	12.2 (0.8)	11.4 (1.5)	16.6 (1.4)	16.3 (1.3)	8.1 (0.7)	4.3 (0.2)
$\tau^2 = 0.05$	27.1 (1.2)	19.0 (1.4)	19.2 (1.3)	**13.8** (0.8)	15.4 (1.1)	15.3 (0.8)	19.1 (1.4)	21.4 (2.9)	12.4 (0.6)	17.3 (0.7)
$\tau^2 = 0.2$	26.7 (1.3)	24.4 (1.1)	24.8 (1.3)	21.7 (0.9)	**20.5** (0.8)	23.2 (1.1)	24.4 (1.1)	24.4 (1.1)	19.4 (0.8)	61.0 (1.9)
$\tau^2 = 0.5$	**26.0** (1.1)	29.3 (1.2)	29.0 (1.4)	27.5 (1.1)	26.0 (1.1)	28.7 (1.2)	29.4 (1.2)	29.4 (1.5)	24.0 (0.9)	124.4 (3.2)
$\epsilon = 0.3$	26.9 (1.0)	20.5 (1.4)	19.2 (1.2)	**17.5** (0.8)	30.6 (1.8)	19.6 (1.3)	24.8 (1.2)	20.6 (1.5)	15.8 (0.8)	17.0 (0.6)
no DP	27.8 (1.2)	19.0 (1.2)	19.4 (1.5)	10.4 (0.3)	**10.1** (0.3)	11.3 (0.5)	11.8 (0.5)	13.8 (0.6)	9.7 (0.3)	16.7 (0.5)
regional*	28.7 (1.2)	26.5 (1.2)	25.3 (1.2)	**20.9** (0.9)	21.2 (0.8)	24.9 (1.1)	26.4 (1.2)	27.1 (1.6)	19.9 (0.7)	48.5 (3.1)
wrong anchor*	34.6 (2.2)	41.9 (2.7)	20.3 (1.5)	16.2 (1.2)	**14.8** (1.1)	21.0 (2.1)	41.8 (4.5)	46.3 (4.3)	15.0 (1.2)	16.9 (0.6)
$\tau^2 = 0.1$, $\epsilon = 0.5$*	26.9 (1.2)	22.8 (1.1)	21.7 (1.3)	**18.4** (0.8)	23.8 (1.5)	21.7 (1.3)	23.9 (1.2)	22.9 (1.1)	17.1 (0.7)	32.7 (1.2)
$K = 48$*	32.4 (1.0)	22.4 (1.6)	22.2 (0.8)	**14.4** (0.5)	14.6 (0.6)	16.0 (0.8)	22.5 (1.3)	24.1 (2.9)	13.2 (0.5)	17.2 (0.4)

in all ten (0.0192 at $\tau^2 = 0.05$), with its worst hospital within a factor 1.08 of its local error in every private setting except $\tau^2 = 0.5$. In worlds where GHB-lite's inversion is exact, per-hospital releases are worth their noise; Section 7 shows that on real hospitals they were not.

Privacy is the binding constraint. GHB-lite cuts the excess log-loss of local estimation by 49% at $\epsilon = 1$ and keeps most of that at $\epsilon = 0.3$ (0.0175 against 0.0273, Simulation 7), whereas an earlier formulation, which released ridge fits and plugged in an estimate of τ^2, kept almost none (0.0259): what a hospital releases, and how it handles τ, matter as much as how much privacy it spends. Joint release among groups of small hospitals helps only when every hospital is small (Simulation 5),

Table 5. Simulation 15: who pays. Left: the largest ratio of any hospital's mean excess log-loss to its own local value (above 1, some hospital is worse off than alone; lowest in bold). Middle: the percentage of hospitals that gain over local estimation. Right: the mean excess log-loss of the eight largest hospitals (10^{-3} nats).

	Worst hospital, relative to local								Hospitals that gain (%)			Eight largest		
Setting	Local	Anchor GD	GHB	GHB-lite	Ditto [Li et al. 2021a]	FedAvg + FT [McMahan et al. 2018]	Decentr. [Lian et al. 2017]	Clust. [Ghosh et al. 2020]	GHB-lite	Ditto [Li et al. 2021a]	FedAvg + FT [McMahan et al. 2018]	Local	GHB-lite	FedAvg + FT [McMahan et al. 2018]
$\tau^2 = 0.01$	1.00	1.77	0.88	**0.79**	0.87	1.87	2.05	1.60	100	100	88	10.3	6.6	9.9
$\tau^2 = 0.05$	1.00	2.10	0.95	**0.85**	0.86	1.39	2.22	1.80	100	100	88	10.0	7.9	9.8
$\tau^2 = 0.2$	1.00	1.56	1.08	1.04	**0.97**	1.43	1.64	1.52	83	100	71	10.1	9.8	11.1
$\tau^2 = 0.5$	**1.00**	2.72	1.32	1.15	1.06	2.58	2.77	2.66	12	46	25	9.5	9.7	15.6
$\epsilon = 0.3$	1.00	2.67	0.96	**0.92**	1.52	2.35	2.10	2.58	100	0	62	9.4	7.9	13.1
no DP	1.00	2.28	0.97	0.87	**0.83**	1.34	1.57	1.25	100	100	92	10.6	6.9	7.9
regional*	1.00	1.69	1.06	1.06	**0.98**	1.52	1.82	1.56	92	100	71	10.0	9.3	11.2
wrong anchor*	1.00	1.57	1.00	0.88	**0.88**	2.96	8.23	2.10	100	100	58	9.6	8.0	14.6
$\tau^2 = 0.1, \epsilon = 0.5$*	1.00	1.26	0.99	**0.97**	1.01	2.35	1.64	1.23	100	92	58	10.2	9.1	16.2
$K = 48$*	1.00	2.81	1.02	0.96	**0.92**	2.39	3.22	2.16	100	100	77	11.2	8.2	12.2

and refits are affordable only under zero-concentrated accounting: after 64 refits a hospital keeps $B = 32.4$ virtual patients, against 1.5 under basic composition (Simulation 14).

6.4 What breaks it

Next, failures on purpose. *Atypical hospitals* (Simulation 9) are the failure Proposition 4.1 predicts. Heavy-tailed effects keep the mean risk low (0.69; 0.65 with a Student-t prior), and an outlying large hospital barely loses (1.11), but an outlying small one loses badly (6.7 times its local risk, at best 4.4 with any safeguard): a small hospital cannot tell from its own data that it is atypical, and only external structure (Simulation 6, Equation (5)) can. *A wrong anchor* (Simulation 11) is harmless when its uncertainty is stated honestly or it is dropped on conflict (at most 0.0167), but an overconfident anchor two standard deviations off raises the error to 0.0352; GHB stays at or below 0.0198 at every error. The conflict check is therefore part of both protocols. *Dishonest hospitals* (Simulation 12) never pushed honest ones below local performance: a large lie inflates $\hat{\tau}^2$ and switches borrowing off (worst 1.00), and screening releases keeps the gain against large lies (0.72) at a small cost when nobody lies.

6.5 Which components and settings matter

Simulation 16 removes one component of GHB-lite at a time, on the same data and noise (Figure 10, Table 6). Every component earns its place somewhere and costs little elsewhere. Releasing a ridge fit instead of the score is worse in seven of eight settings, by up to 1.39; releasing $X^\top y$ is 1.74 times worse under case-mix shift, which the score at the anchor absorbs; inverse-variance weighting is worth up to 1.86, the anchor up to 1.90, and the conflict check 1.36 when the anchor is wrong. The rule for τ is the most delicate: plug-in rules win slightly in a few settings and lose by up to 1.37 (posterior mean) or 1.33 (DerSimonian-Laird [DerSimonian and Laird 1986]) elsewhere, while integrating τ out with each hospital's evidence is within 4% of the best rule everywhere. Gossip and zero-concentrated accounting cost nothing for a single release.

Simulation 17 varies GHB-lite's own settings and the network around it (Figures 11 and 12, Tables 7 and 8). Most settings hardly matter: the number of quantiles from three, the public sample from 2,000 draws, the conflict threshold from 2.5, gossip from 20 rounds. Two matter: the half-Cauchy scale trades nearly identical against heterogeneous hospitals (0.90 to 1.17), with the default 0.5 within 11% of the best everywhere, and an overstated anchor uncertainty costs up to 1.41, where the anchor carries most of the information; a tighter default prior also helps in this world (0.90), whose coefficients are small, but we keep the generic 2.5. Across 6 to 96 hospitals, median sizes of 25 to 400, budgets from $\epsilon = 0.1$ to none, and every topology, the error stays between 0.40 and 0.69 of local and the worst hospital within a factor 1.03; the gain grows with the budget and with the number of hospitals up to about 48, and is largest for hospitals of about 100 patients.

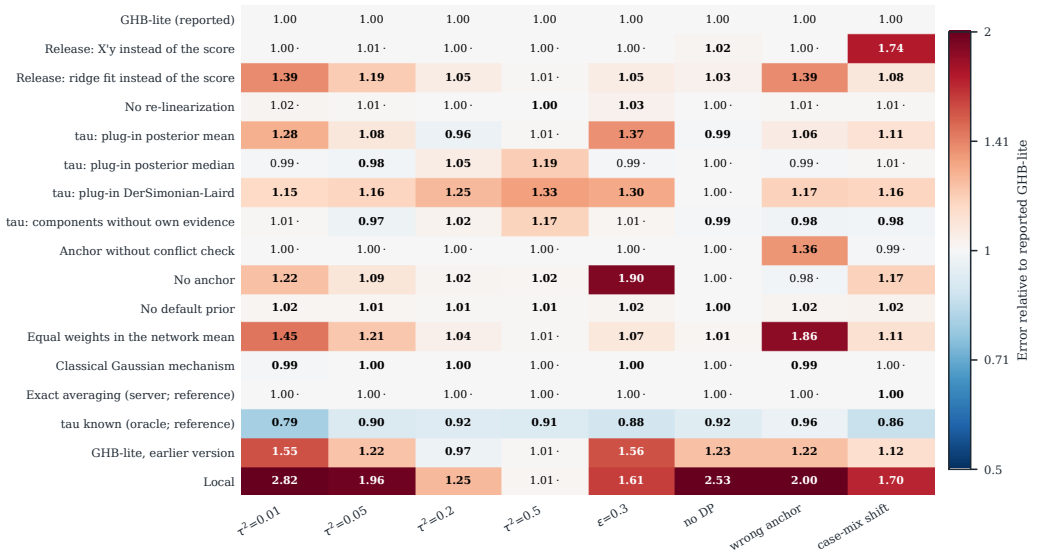


Fig. 10. Simulation 16: ablation of GHB-lite (logistic heads, 24 hospitals, 80 replicates per setting). Each cell is the mean excess log-loss of a variant divided by that of GHB-lite as reported, computed on the same data and noise; above 1 (red), removing or replacing the component hurts. Bold: paired difference larger than two standard errors; a trailing dot marks differences within Monte Carlo error. The last three rows are references.

Table 6. Simulation 16: exact values behind Figure 10. First row: mean excess log-loss of GHB-lite as reported (10^{-3} nats). Other rows: the mean excess log-loss of each variant divided by that of GHB-lite on the same data and noise (80 replicates per setting); bold where the paired difference exceeds two standard errors. The last rows are references.

Variant	$\tau^2 = 0.01$	$\tau^2 = 0.05$	$\tau^2 = 0.2$	$\tau^2 = 0.5$	$\epsilon = 0.3$	no DP	wrong anchor	shift
Reported GHB-lite (10^{-3} nats)	9.8	14.0	21.6	27.6	17.1	11.1	16.1	13.5
Release: $X^\top y$ instead of the score	1.00	1.01	1.00	1.00	1.00	**1.02**	1.00	**1.74**
Release: ridge fit instead of the score	**1.39**	**1.19**	**1.05**	1.01	**1.05**	**1.03**	**1.39**	**1.08**
No re-linearization	1.02	1.01	1.00	**1.00**	**1.03**	1.00	1.01	1.01
τ: plug-in posterior mean	**1.28**	**1.08**	**0.96**	1.01	**1.37**	**0.99**	**1.06**	**1.11**
τ: plug-in posterior median	0.99	**0.98**	**1.05**	**1.19**	0.99	1.00	0.99	1.01
τ: plug-in DerSimonian-Laird [DerSimonian and Laird 1986]	**1.15**	**1.16**	**1.25**	**1.33**	**1.30**	1.00	**1.17**	**1.16**
τ: components without own evidence	1.01	**0.97**	**1.02**	**1.17**	1.01	**0.99**	**0.98**	**0.98**
Anchor without conflict check	1.00	1.00	1.00	1.00	1.00	1.00	**1.36**	0.99
No anchor	**1.22**	**1.09**	**1.02**	**1.02**	**1.90**	1.00	0.98	**1.17**
No default prior	**1.02**	**1.01**	**1.01**	**1.01**	**1.02**	**1.00**	**1.02**	**1.02**
Equal weights in the network mean	**1.45**	**1.21**	**1.04**	1.01	**1.07**	**1.01**	**1.86**	**1.11**
Classical Gaussian mechanism	**0.99**	**1.00**	**1.00**	1.00	**1.00**	1.00	**0.99**	1.00
Exact averaging (server; reference)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	**1.00**
τ known (oracle; reference)	**0.79**	**0.90**	**0.92**	**0.91**	**0.88**	**0.92**	**0.96**	**0.86**
GHB-lite, earlier version	**1.55**	**1.22**	**0.97**	1.01	**1.56**	**1.23**	**1.22**	**1.12**
Local	**2.82**	**1.96**	**1.25**	1.01	**1.61**	**2.53**	**2.00**	**1.70**

The same analysis for GHB. Simulation 16B repeats the ablation and sensitivity analysis for GHB in the same settings, on paired data and noise (Figure 13; 40 replicates per setting; supplement). In this world, which rewards borrowing, GHB's cautious choices cost what they protect against

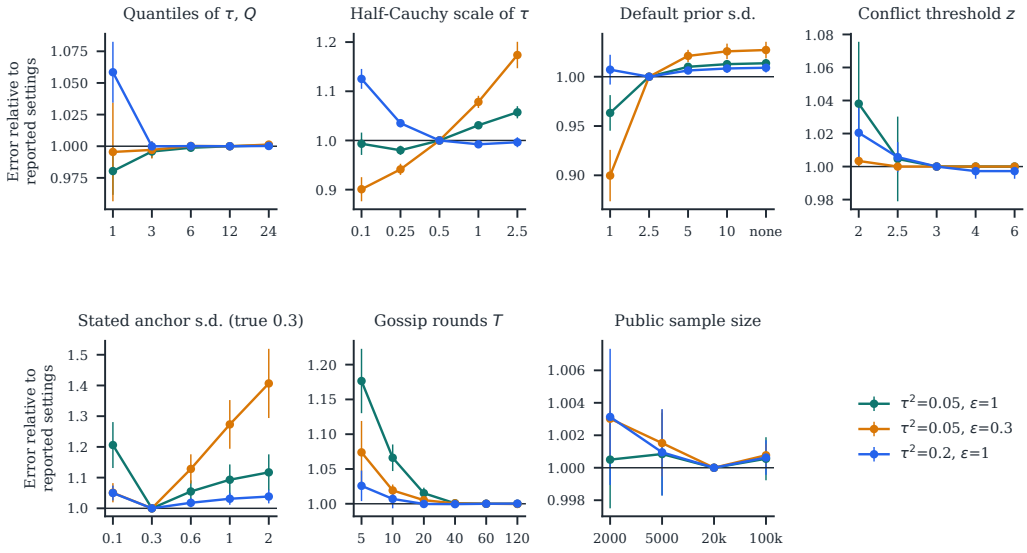


Fig. 11. Simulation 17, part A: sensitivity of GHB-lite to its own settings (logistic heads, 40 paired replicates per point). Each point is the mean excess log-loss with the setting changed, divided by that with the reported settings ($Q = 12$, scale 0.5, default prior s.d. 2.5, $z = 3$, stated anchor s.d. 0.3, $T = 60$, 20,000 public draws). Bars: ±2 standard errors of the paired ratio.

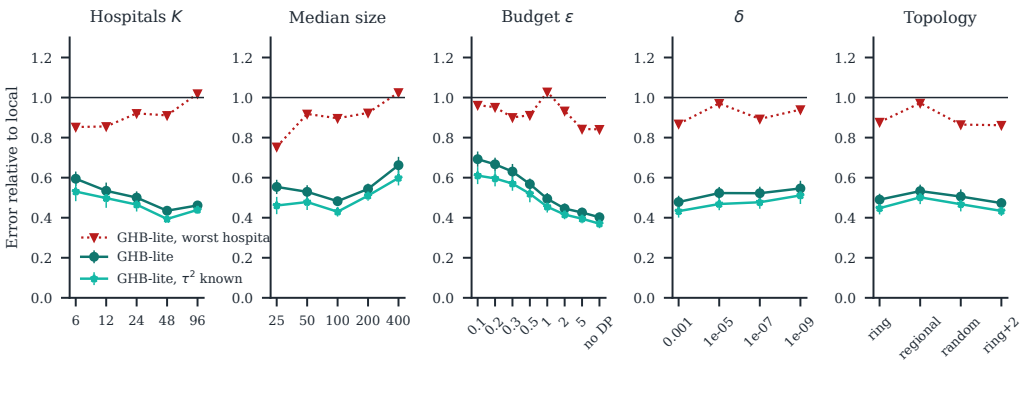


Fig. 12. Simulation 17, part B: GHB-lite across network conditions ($\tau^2 = 0.05$, $\epsilon = 1$, 24 hospitals of median size 100 on the ring with two-hop links unless varied; 40 replicates per point). Mean excess log-loss relative to local estimation for GHB-lite and for GHB-lite with τ^2 known (±2 standard errors), and the largest ratio of any hospital's error to its own local error (dotted).

on real hospitals: borrowing the intercept would lower its error to as little as 0.72 of the released version's, dropping the non-borrowing components to 0.93, and a wider prior on τ raises it by up to 1.24; the propagated privacy covariance, the leave-one-out step, the tight sensitivity and three or more quantiles move it by 1% or less, while heavier non-borrowing weight raises it by up to 1.24 and a single quantile by up to 1.14. Across the conditions of Simulation 17 its error stays

Table 7. Simulation 17, part A: exact values behind Figure 11. Mean excess log-loss with one setting changed, divided by that with the reported settings (italic), on the same data and releases; two standard errors of the paired ratio in grey (40 replicates).

Setting	Value	$\tau^2 = 0.05$, $\epsilon = 1$	$\tau^2 = 0.05$, $\epsilon = 0.3$	$\tau^2 = 0.2$, $\epsilon = 1$
Quantiles of τ, Q	1	0.980 (0.019)	0.995 (0.039)	1.058 (0.024)
	3	0.996 (0.005)	0.997 (0.007)	1.000 (0.004)
	6	0.999 (0.002)	1.000 (0.002)	1.000 (0.002)
	12	1	1	1
	24	1.001 (0.001)	1.001 (0.002)	1.000 (0.001)
Half-Cauchy scale of τ	0.1	0.993 (0.023)	0.901 (0.025)	1.125 (0.020)
	0.25	0.980 (0.009)	0.941 (0.011)	1.035 (0.008)
	0.5	1	1	1
	1.0	1.031 (0.007)	1.078 (0.012)	0.992 (0.006)
	2.5	1.057 (0.012)	1.174 (0.027)	0.996 (0.010)
Default prior s.d.	1.0	0.963 (0.018)	0.900 (0.026)	1.007 (0.015)
	2.5	1	1	1
	5.0	1.010 (0.003)	1.021 (0.006)	1.006 (0.003)
	10.0	1.013 (0.004)	1.026 (0.008)	1.008 (0.004)
	none	1.014 (0.005)	1.027 (0.008)	1.009 (0.005)
Conflict threshold z	2.0	1.038 (0.037)	1.003 (0.003)	1.020 (0.018)
	2.5	1.005 (0.026)	1.000 (0.000)	1.006 (0.009)
	3.0	1	1	1
	4.0	1.000 (0.000)	1.000 (0.000)	0.997 (0.005)
	6.0	1.000 (0.000)	1.000 (0.000)	0.997 (0.005)
Stated anchor s.d.	0.1	1.206 (0.075)	1.051 (0.031)	1.050 (0.024)
	0.3	1	1	1
	0.6	1.055 (0.036)	1.128 (0.048)	1.018 (0.014)
	1.0	1.093 (0.050)	1.273 (0.079)	1.031 (0.020)
	2.0	1.117 (0.059)	1.407 (0.113)	1.038 (0.022)
Gossip rounds T	5	1.176 (0.046)	1.074 (0.045)	1.026 (0.022)
	10	1.066 (0.019)	1.019 (0.008)	1.007 (0.014)
	20	1.015 (0.008)	1.005 (0.003)	1.000 (0.005)
	40	1.001 (0.002)	1.001 (0.001)	0.999 (0.001)
	60	1	1	1
	120	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)
Public sample size	2000	1.001 (0.003)	1.003 (0.002)	1.003 (0.004)
	5000	1.001 (0.003)	1.002 (0.002)	1.001 (0.003)
	20000	1	1	1
	100000	1.001 (0.001)	1.001 (0.001)	1.001 (0.001)

between 0.64 and 0.82 of local estimation, with its worst hospital within a factor 1.11, always behind GHB-lite in this world. Section 7.2 shows which choices pay on real hospitals.

6.6 Fairness to groups of patients

Every simulation so far asks whether some hospital pays for the others; the last asks whether some group of patients does. In Simulation 18 one feature becomes a protected attribute S whose share varies across hospitals, and its coefficient is each hospital's true disparity, shared, varying, rare or of either sign across hospitals. A variant, *GHB-lite, S local*, keeps the disparity coefficient unborrowed at no privacy cost. FAIR-FATE [Salazar et al. 2023], with its paper's settings, and federated averaging [McMahan et al. 2017] are references: both use a server, FAIR-FATE also a labelled validation set, and neither is private (Figure 14, Table 9).

GHB-lite or its variant is the most accurate private method for the minority in every scenario, and the most accurate of all methods whenever hospitals disagree about the disparity. It roughly halves the minority's excess log-loss of local estimation when the disparity is shared (17.8 against 34.3 $\times 10^{-3}$ nats); when disparities vary, single models fit the minority badly (global 60.7, FAIR-FATE

Table 8. Simulation 17, part B: exact values behind Figure 12. Mean excess log-loss of GHB-lite relative to local estimation (two standard errors in grey), the same for GHB-lite with τ^2 known, the largest ratio of any hospital's error to its own local error, and the absolute mean excess log-loss of GHB-lite and of local estimation (10^{-3} nats). $\tau^2 = 0.05$, $\epsilon = 1$, $\delta = 10^{-5}$, 24 hospitals of median size 100 on the ring with two-hop links unless varied.

Condition	Value	GHB-lite / local	GHB-lite, τ^2 known / local	Worst hospital	GHB-lite	Local
Hospitals K	6	0.595 (0.036)	0.531	0.85	13.7	23.0
	12	0.534 (0.041)	0.498	0.86	13.4	25.2
	24	0.500 (0.031)	0.466	0.92	13.7	27.4
	48	0.435 (0.018)	0.394	0.91	14.0	32.2
	96	0.461 (0.020)	0.439	1.02	13.2	28.5
Median size	25	0.554 (0.036)	0.461	0.75	33.9	61.2
	50	0.529 (0.033)	0.478	0.92	23.0	43.5
	100	0.482 (0.022)	0.430	0.90	13.3	27.6
	200	0.544 (0.026)	0.510	0.92	8.7	16.1
	400	0.662 (0.042)	0.599	1.03	5.6	8.4
Budget ϵ	0.1	0.692 (0.038)	0.610	0.96	18.9	27.3
	0.2	0.667 (0.033)	0.596	0.95	18.2	27.3
	0.3	0.630 (0.038)	0.570	0.90	16.4	26.0
	0.5	0.568 (0.028)	0.520	0.91	16.7	29.4
	1.0	0.495 (0.029)	0.455	1.03	13.8	27.8
	2.0	0.446 (0.025)	0.416	0.93	12.1	27.3
	5.0	0.426 (0.023)	0.395	0.84	11.1	26.1
	no DP	0.402 (0.023)	0.370	0.84	10.9	27.2
δ	0.001	0.478 (0.031)	0.433	0.87	13.0	27.2
	1e-05	0.523 (0.031)	0.468	0.97	14.5	27.8
	1e-07	0.522 (0.033)	0.477	0.89	14.2	27.2
	1e-09	0.546 (0.038)	0.511	0.94	14.9	27.2
Topology	ring	0.490 (0.029)	0.448	0.88	13.9	28.4
	regional	0.533 (0.032)	0.502	0.97	14.2	26.7
	sparse random	0.506 (0.036)	0.467	0.86	13.6	27.0
	ring, two-hop	0.473 (0.025)	0.434	0.86	13.1	27.7

72.1) while GHB-lite reaches 27.3. Because it learns τ coefficient by coefficient, GHB-lite borrows a disparity hospitals agree on and barely borrows one they disagree on, so keeping it local is a safeguard, not a default. GHB-lite leaves fairness ratios where the true models put them (within 0.02), while FAIR-FATE moves them toward parity, which, when the disparity is real, distorts the minority's predicted risks (30.3 against 17.8). Calibrated personalized risk and parity of decisions are different goals, and the choice belongs to the deployment.

7 REAL HOSPITAL NETWORKS

The simulations gave GHB-lite every advantage: an exact public feature distribution and informative per-hospital releases. Real hospitals give neither. This section follows what happened on them. Small real networks show why per-hospital releases fail and why GHB releases sums (Section 7.1); eight held-out cancer networks measure GHB against the federated and personalized literature (Section 7.2); the FLamby benchmark and two image tasks test how far the result travels (Sections 7.4 and 7.5).

All studies share one protocol. We treat $\epsilon = 1$ as the budget a network should use and $\epsilon = 10$ as a weak-privacy reference, whose guarantee is loose (Section 7.6). Every hospital is scored on its own held-out patients, errors are relative to local estimation under the default prior $\mathcal{N}(0, 2.5^2)$, and every private method spends the same budget, $\epsilon \in \{1, 10\}$ with $\delta = 10^{-5}$, under zero-concentrated accounting. The literature methods have grids of hyperparameters, and how they are tuned decides

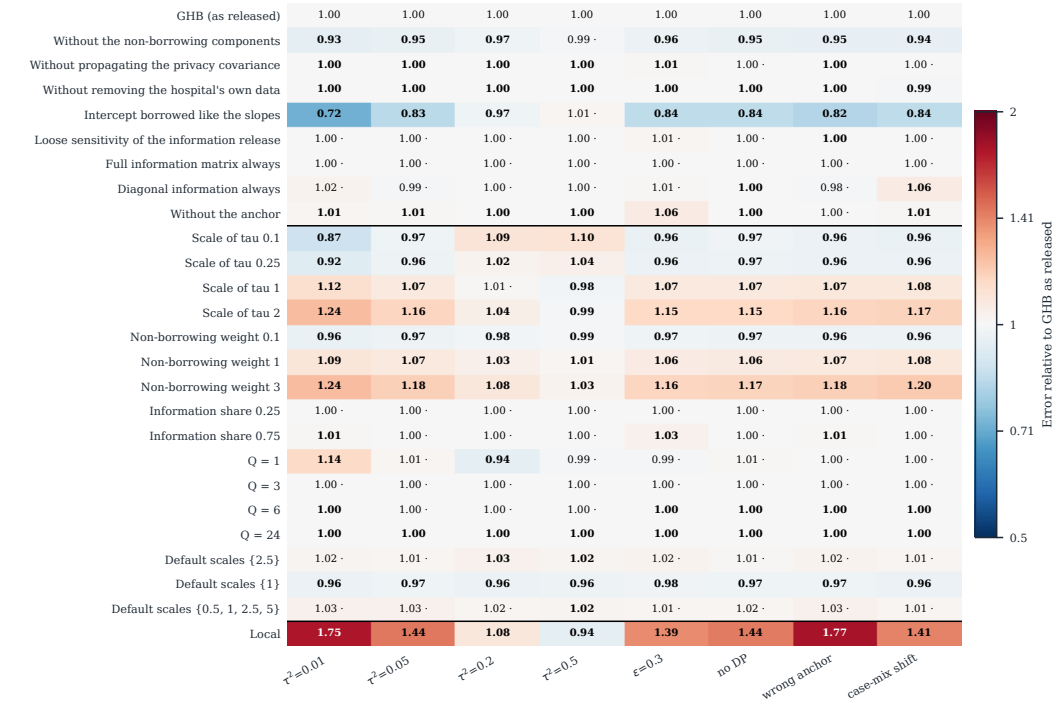


Fig. 13. Simulation 16B: ablation (top block) and sensitivity (middle block) of GHB as released (logistic heads, 24 hospitals, 40 replicates per setting). Each cell is the mean excess log-loss of a variant divided by that of GHB on the same data and noise; above 1 (red), the change hurts. Bold: paired difference larger than two standard errors; a trailing dot marks differences within Monte Carlo error. Last row: local estimation, a reference.

how they compare, so two arms are reported. In the *realistic* arm each method uses the configuration that is best on other networks, which a consortium could do; in the *oracle* arm it uses the configuration that is best on the test patients themselves, which no hospital can do. GHB and GHB-lite tune nothing. pFedMe [T. Dinh et al. 2020] releases the change of a proximal fit whose sensitivity is $S\beta\, 2G/\lambda$ for S local steps of size β; we use this calibration, under which its updates are mostly noise at strong privacy.

7.1 Small networks: why the releases are sums

Networks. Seven public networks with a few dozen patients per hospital: Arizona Medicare [Hilbe 2011] (1,466 cardiovascular admissions at 43 hospitals, in-hospital death, four predictors) and six TCGA cancer types [Cancer Genome Atlas Research Network et al. 2013; Grossman et al. 2016] (lung squamous, head and neck, bladder, stomach, liver and clear-cell kidney; 7 to 17 hospitals; death within two years; age, sex and stage). UCI Heart Disease [Detrano et al. 1989] served to develop GHB-lite and to tune the baselines in a third arm. Twelve literature methods are compared: DP-FedAvg [McMahan et al. 2018] alone, with fine-tuning, and with fine-tuning and secure aggregation [Bonawitz et al. 2017]; decentralized DP-SGD [Lian et al. 2017]; clustered DP-FedAvg [Ghosh et al. 2020]; a one-shot Ditto-style model [Li et al. 2021a]; Ditto [Li et al. 2021a], pFedMe [T. Dinh et al. 2020], APFL [Deng et al. 2020], Per-FedAvg [Fallah et al. 2020], FedRep [Collins et al. 2021] and FedProx [Li et al. 2020]. Here the realistic arm is leave-one-network-out: each method uses the

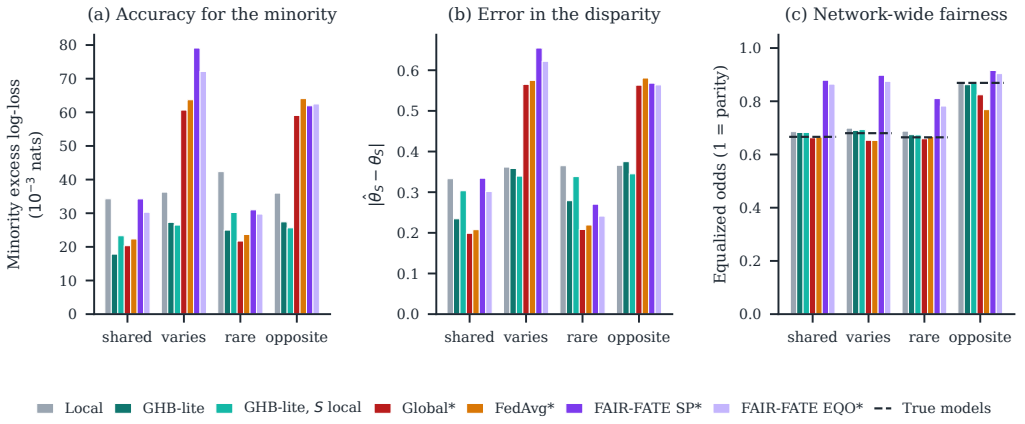


Fig. 14. Simulation 18: a protected attribute S ($\epsilon = 1$, 24 hospitals, 100 replicates per scenario). (a) Mean excess log-loss of the minority group. (b) Mean absolute error of the estimated disparity, the coefficient of S. (c) Equalized odds of the network's decisions at threshold 0.5 (1 is parity); dashed: the true models. $*$: not private, with a server; FAIR-FATE [Salazar et al. 2023] also uses a labelled validation set at the server.

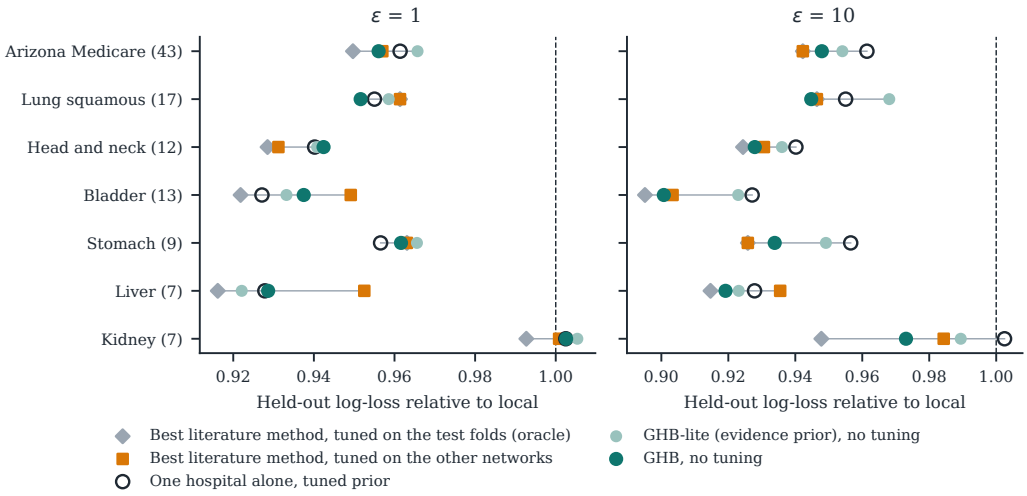


Fig. 15. Seven real networks (hospitals in parentheses): held-out log-loss relative to local estimation, at $\epsilon = 1$ and $\epsilon = 10$. Dark teal: GHB; light teal: GHB-lite with the evidence-weighted prior; neither is tuned. Orange: the best of twelve federated and personalized methods for that network, each tuned on the other six networks and UCI Heart Disease. Grey: the same, tuned on the network's own held-out folds (oracle). Open circle: one hospital alone with a tuned prior. Choosing the best of twelve methods separately for every network is itself an advantage no hospital has.

configuration with the lowest mean error on the other networks. These networks were also used in designing GHB, so they show what it does, not that it generalizes; Section 7.2 provides that test.

Results. At $\epsilon = 1$ collaboration adds little for anyone (Figure 15, Table 10). GHB reaches 0.954 of the local error, no literature method tuned on other networks does better on average (Ditto [Li et al.

Table 9. Simulation 18: exact values behind Figure 14 (100 replicates per scenario). Mean excess log-loss over all patients, the minority ($S = 1$) and the majority (10^{-3} nats; two standard errors in grey); mean absolute error of the estimated disparity $|\hat{\theta}_S - \theta_S|$; mean absolute calibration bias in the minority, $|\text{mean}(\hat{p} - p \mid S = 1)|$ (percentage points); the largest ratio of any hospital's excess log-loss to its local one (Worst); and the statistical parity, equal opportunity and equalized odds ratios of the network's decisions at threshold 0.5 (1 is parity). Bold: best of the private methods. †: not private, one model trained with a server. FAIR-FATE follows Algorithm 1 of [Salazar et al. 2023]: a third of the hospitals per round, 10 local epochs of mini-batch gradient descent (batches of 10, learning rate 0.01), $\lambda_0 = 0.5$, $\rho = 0.05$, MAX = 0.9, $\beta_0 = 0.9$, 100 rounds, and a labelled validation set of a fifth of all patients at the server; federated averaging [McMahan et al. 2017] uses the same protocol without the fairness update. All federated models start from the anchor μ_0.

	Excess log-loss						Fairness of decisions		
Method	All	Minority	Majority	$\lvert\hat{\theta}_S - \theta_S\rvert$	Calib. bias	Worst	SP	EO	EQO
Shared disparity ($\tau_S^2 = 0.05$)									
Local	25.4 (0.8)	34.3 (1.4)	23.3	0.333	6.3	1.00	0.65	0.76	0.69
GHB-lite	13.2 (0.5)	**17.8** (0.8)	11.7	**0.234**	4.9	0.86	0.65	0.76	0.68
GHB-lite, S local	14.4 (0.5)	23.3 (1.1)	11.9	0.304	6.0	0.91	0.65	0.77	0.68
Global†	16.1 (0.5)	20.3 (0.8)	14.5	0.199	5.3	2.72	0.63	0.74	0.66
FedAvg† [McMahan et al. 2017]	17.7 (0.6)	22.3 (1.1)	16.0	0.208	5.5	2.56	0.63	0.74	0.67
FAIR-FATE, SP† [Salazar et al. 2023]	24.2 (1.3)	34.3 (2.5)	19.9	0.334	7.5	4.47	0.83	0.91	0.88
FAIR-FATE, EQO† [Salazar et al. 2023]	22.1 (0.9)	30.3 (1.8)	18.7	0.302	6.9	4.11	0.81	0.90	0.86
True models							0.63	0.75	0.67
Disparity varies ($\tau_S^2 = 0.5$)									
Local	25.9 (0.9)	36.3 (1.7)	23.7	0.362	6.4	1.00	0.65	0.74	0.70
GHB-lite	15.8 (0.6)	27.3 (1.4)	12.9	0.358	6.5	0.98	0.65	0.74	0.69
GHB-lite, S local	15.5 (0.6)	**26.5** (1.4)	12.7	**0.340**	6.3	0.96	0.65	0.74	0.69
Global†	28.7 (1.1)	60.7 (2.8)	15.3	0.565	11.7	4.83	0.62	0.75	0.65
FedAvg† [McMahan et al. 2017]	30.6 (1.2)	63.7 (3.2)	16.4	0.575	12.0	4.52	0.63	0.74	0.65
FAIR-FATE, SP† [Salazar et al. 2023]	37.1 (1.7)	79.1 (4.8)	18.7	0.655	13.4	7.62	0.89	0.92	0.90
FAIR-FATE, EQO† [Salazar et al. 2023]	34.5 (1.4)	72.1 (3.7)	18.3	0.622	12.7	7.03	0.84	0.92	0.87
True models							0.63	0.72	0.68
Rare minority (share 0.1)									
Local	24.1 (0.8)	42.4 (1.8)	22.7	0.365	7.8	1.00	0.65	0.77	0.69
GHB-lite	13.0 (0.5)	**25.0** (2.0)	11.9	**0.279**	6.2	0.84	0.64	0.76	0.67
GHB-lite, S local	13.5 (0.5)	30.2 (1.6)	12.0	0.338	7.3	0.88	0.64	0.76	0.67
Global†	15.8 (0.5)	21.7 (1.0)	15.1	0.208	5.4	2.37	0.62	0.74	0.66
FedAvg† [McMahan et al. 2017]	17.3 (0.7)	23.7 (1.2)	16.5	0.219	5.6	2.30	0.63	0.75	0.67
FAIR-FATE, SP† [Salazar et al. 2023]	23.3 (1.6)	31.0 (2.5)	22.3	0.270	6.4	4.10	0.76	0.86	0.81
FAIR-FATE, EQO† [Salazar et al. 2023]	23.0 (1.5)	29.8 (2.4)	22.2	0.241	6.2	3.95	0.74	0.84	0.78
True models							0.63	0.75	0.66
Opposite disparities ($\mu_S = 0$, $\tau_S^2 = 0.5$)									
Local	26.1 (0.9)	36.0 (1.8)	24.1	0.366	6.4	1.00	0.87	0.89	0.87
GHB-lite	16.0 (0.7)	27.4 (2.6)	13.2	0.375	6.5	0.94	0.86	0.89	0.86
GHB-lite, S local	15.3 (0.5)	**25.7** (1.5)	12.7	**0.345**	6.2	0.90	0.86	0.89	0.87
Global†	28.5 (1.0)	59.1 (2.8)	15.7	0.563	11.4	4.09	0.82	0.86	0.82
FedAvg† [McMahan et al. 2017]	31.4 (1.2)	64.1 (3.2)	17.5	0.581	11.8	4.07	0.77	0.81	0.77
FAIR-FATE, SP† [Salazar et al. 2023]	31.3 (1.3)	61.9 (2.8)	17.8	0.568	11.5	5.76	0.92	0.93	0.92
FAIR-FATE, EQO† [Salazar et al. 2023]	32.4 (1.4)	62.5 (2.7)	19.0	0.564	11.5	5.86	0.91	0.93	0.90
True models							0.86	0.88	0.87

2021a] comes closest, 0.962), and only a few tuned on the test folds do (best: DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation [Bonawitz et al. 2017], 0.949). But a single hospital that regularizes well matches every collaborative method: local estimation with a tuned prior reaches 0.953. At $\epsilon = 10$ collaboration starts to pay. GHB (0.935) beats local estimation with a tuned prior in 7 of the seven networks, and no method a consortium could tune is more accurate on average (DP-FedAvg with fine-tuning and secure aggregation 0.946, Ditto 0.949); tuned on the test

Table 10. Seven real networks: held-out log-loss relative to local estimation, mean over the networks, for every method in the oracle arm (tuned on the network's own held-out folds) and the leave-one-network-out arm (tuned on the other networks); Bold: the lowest mean per column among GHB, GHB-lite and the federated and personalized methods. Wins: in how many of the seven networks GHB, untuned, has a lower error than that method in the leave-one-network-out arm. ‡: no other hospital. Every method spends the same budget. Per-network values are in the supplement.

	$\epsilon = 1$			$\epsilon = 10$		
Method	Oracle	Other networks	Wins	Oracle	Other networks	Wins
GHB and the local reference						
GHB, no tuning	0.954	0.954		0.935	**0.935**	
GHB-lite, no tuning	1.000	1.000		0.964	0.964	
GHB-lite, evidence prior, no tuning	0.956	0.956		0.949	0.949	
GHB-lite with secure aggregation, no tuning	0.982	0.982		0.950	0.950	
GHB-lite, tuned prior	0.951	**0.953**		0.948	0.950	
Local, tuned prior‡	0.951	0.953		0.951	0.953	
Federated and personalized methods (same budget)						
Ditto [Li et al. 2021a]	0.962	0.962	6	0.944	0.949	7
DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	**0.949**	0.985	6	**0.928**	0.946	5
APFL [Deng et al. 2020]	0.958	0.988	7	0.944	0.963	7
DP-FedAvg + FT [McMahan et al. 2018]	0.951	0.994	7	0.940	0.952	7
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	0.954	0.997	7	0.948	0.967	7
Decentralized DP-SGD [Lian et al. 2017]	0.972	1.009	7	0.947	0.964	7
FedProx [Li et al. 2020]	1.006	1.013	7	0.942	0.958	7
FedRep [Collins et al. 2021]	1.045	1.046	7	1.013	1.023	7
DP-FedAvg [McMahan et al. 2018]	1.085	1.112	7	1.029	1.044	7
Per-FedAvg [Fallah et al. 2020]	1.161	1.165	7	1.009	1.025	7
Ditto-style (one-shot) [Li et al. 2021a]	1.321	1.333	7	0.984	1.015	7
pFedMe [T. Dinh et al. 2020]	7.838	8.052	7	1.119	1.132	7

folds, DP-FedAvg with fine-tuning and secure aggregation leads by less than 1% (0.928). Tuning is most of the baselines' advantage: tuned on UCI Heart Disease alone, DP-FedAvg with fine-tuning and secure aggregation and Ditto fall to 1.022 and 1.045 at $\epsilon = 1$, and FedRep [Collins et al. 2021] diverges.

GHB-lite, in contrast, stays at the local level (1.000 at $\epsilon = 1$). Integrating its prior scale out by each hospital's evidence (GHB-lite with the evidence prior, 0.956) helps only as much as a better-regularized local model, and secure aggregation of per-hospital quantities, still read through a public feature distribution, adds nothing further. As long as the network estimate passes through a public feature distribution and per-hospital noise, it carries too little to beat a well-regularized hospital. That is the reason GHB releases network sums of the hospitals' real scores and information (Section 5.2).

7.1.1 How much the size of a hospital matters. UCI Heart Disease shows the mechanism most clearly, because each hospital's size can be varied by subsampling. Its four hospitals, Cleveland (303 patients), Budapest (294), Zurich and Basel (123) and Long Beach (200), refer patients for coronary angiography, with disease prevalence from 36% to 93%. Each fits the pretest-probability model (age scaled by public constants, sex and chest-pain type, $\|x\| \leq \sqrt{5}$), trained on at most 20, 40 or 80 patients or on all, and scored on its own held-out fold (100 splits). GHB-lite inverts its releases with a nominal chest-pain clinic population, not the data; the baselines are those of Simulation 15, tuned on the held-out folds.

At $\epsilon = 1$ no private method gains from collaboration (Figure 16, Table 11). GHB-lite stays within 1% of local estimation at every size, with its worst hospital within a factor 1.01. The reason is the noise: a hospital of 40 patients releases noise about 26 times its own sampling error, and with three others to average over, M dwarfs τ^2 and Equation (3) leaves nothing to borrow; GHB-lite

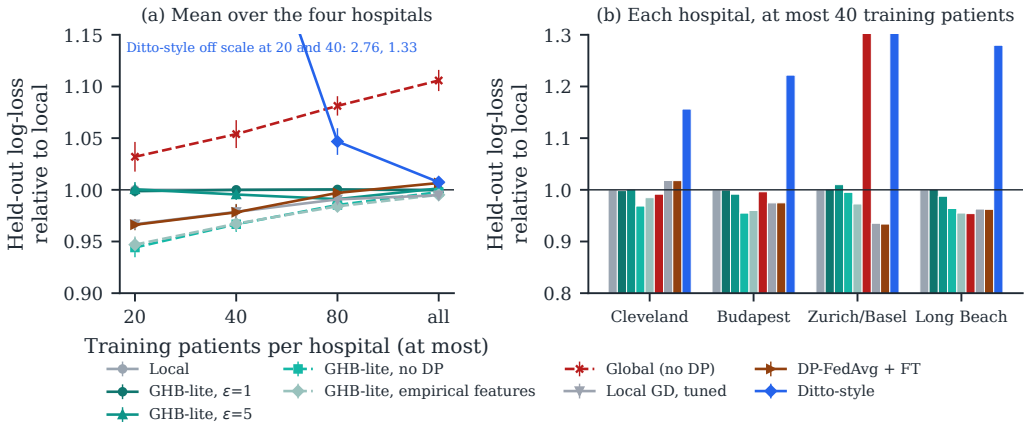


Fig. 16. UCI Heart Disease, four hospitals: held-out log-loss relative to local estimation (repeated stratified five-fold cross-validation, 100 splits; ±2 paired standard errors). (a) Mean over hospitals against the number of training patients per hospital. (b) Each hospital, with at most 40 training patients (bars clipped at 1.3; values in Table 11 and the supplement). Dashed: without privacy. The baselines' hyperparameters are tuned on the held-out folds.

Table 11. UCI Heart Disease, GHB-lite: held-out log-loss relative to local estimation (two paired standard errors in grey) and the largest ratio of any hospital's held-out log-loss to its local one (Worst), by the number of training patients per hospital (100 splits). First row: local log-loss (nats). DP-FedAvg [McMahan et al. 2018] + FT, Ditto-style [Li et al. 2021a] and local GD are tuned on the held-out folds. Bold: the lowest per column among the private collaborative methods (local, global and non-private rows excluded). †: not private.

	≤ 20 patients		≤ 40		≤ 80		All	
Method	Relative	Worst	Relative	Worst	Relative	Worst	Relative	Worst
Local log-loss	0.483		0.460		0.443		0.434	
Local	1.000 (0.000)	1.00	1.000 (0.000)	1.00	1.000 (0.000)	1.00	1.000 (0.000)	1.00
GHB-lite, $\epsilon = 1$	0.999 (0.001)	1.00	1.000 (0.002)	**1.00**	1.000 (0.002)	1.01	**1.000** (0.002)	**1.00**
GHB-lite, $\epsilon = 5$	1.000 (0.007)	1.01	0.995 (0.005)	1.01	**0.991** (0.004)	**1.00**	1.001 (0.004)	1.03
DP-FedAvg + FT, $\epsilon = 1$ [McMahan et al. 2018]	**0.966** (0.005)	**0.99**	**0.978** (0.008)	1.02	0.997 (0.006)	1.03	1.007 (0.005)	1.05
Ditto-style, $\epsilon = 1$ [Li et al. 2021a]	2.765 (0.334)	5.12	1.327 (0.054)	2.06	1.047 (0.013)	1.16	1.007 (0.004)	1.04
Local GD, tuned stopping	0.967 (0.005)	0.99	0.979 (0.008)	1.02	0.991 (0.002)	1.00	0.995 (0.002)	1.00
GHB-lite, no DP†	0.944 (0.009)	1.02	0.967 (0.007)	1.00	0.985 (0.005)	1.01	0.998 (0.004)	1.02
GHB-lite, empirical features†	0.947 (0.008)	0.99	0.967 (0.007)	0.98	0.984 (0.004)	1.00	0.995 (0.002)	1.00
Global†	1.032 (0.014)	1.58	1.054 (0.013)	1.57	1.081 (0.009)	1.59	1.106 (0.010)	1.81

detects this and falls back to local estimation without tuning. DP-FedAvg with fine-tuning appears to gain (0.97), but a local control gains exactly as much (0.97): its configuration is one private round and 30 local steps, so its gain is early stopping tuned on the test folds. Without privacy GHB-lite borrows, 0.94 of the local error with 20 patients, fading to 1.00 with all of them, as Equation (3) predicts, while a pooled model makes the Swiss hospital 1.57 to 1.81 times worse off. Under privacy, collaboration needs more hospitals, a larger budget, or noise calibrated to the network rather than to each hospital.

Table 12. Five TCGA networks held out from GHB's development (hospitals: BRCA 8, MESO 9, OV 15, UCEC 15, UCS 5): held-out log-loss relative to local estimation under the default prior (ten repetitions of five-fold cross-validation). GHB and GHB-lite have nothing to tune. Every literature method uses the configuration that is best on the twelve development networks, fixed before these runs. Bold: the lowest per network among methods that use other hospitals, oracle excluded. Wins: networks in which GHB is more accurate. ‡: no other hospital. ⋆: for each network, the best configuration of the best of the six methods on that network's own held-out folds, which no hospital can do.

Method	BRCA	MESO	OV	UCEC	UCS	Mean	Wins
$\epsilon = 1$							
GHB, no tuning	**0.879**	0.909	0.979	**0.957**	0.899	**0.925**	
GHB-lite (evidence prior), no tuning	0.901	**0.893**	0.982	0.988	**0.895**	0.932	3/5
Local, evidence prior‡	0.902	0.895	0.983	0.992	0.894	0.933	3/5
DP-FedAvg + FT, tuned on development [McMahan et al. 2018]	0.898	0.952	1.004	1.108	0.921	0.976	5/5
DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018]	0.911	1.027	0.999	1.070	1.019	1.005	5/5
APFL, tuned on development [Deng et al. 2020]	0.918	0.987	1.010	1.134	0.986	1.007	5/5
Ditto, tuned on development [Li et al. 2021a]	0.967	1.503	**0.978**	0.997	1.556	1.200	4/5
pFedMe, tuned on development [T. Dinh et al. 2020]	13.333	15.218	7.600	9.511	12.970	11.726	5/5
Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018]	0.926	0.989	1.007	1.112	0.965	1.000	5/5
Best of the six, oracle per network⋆	0.895	0.914	0.978	0.962	0.917	0.933	4/5
$\epsilon = 10$							
GHB, no tuning	**0.859**	0.897	**0.967**	**0.932**	0.887	**0.909**	
GHB-lite (evidence prior), no tuning	0.948	0.958	1.005	1.005	0.907	0.965	5/5
Local, evidence prior‡	0.902	0.895	0.983	0.992	0.894	0.933	4/5
DP-FedAvg + FT, tuned on development [McMahan et al. 2018]	0.889	0.947	0.997	1.061	0.906	0.960	5/5
DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018]	0.903	1.008	0.975	0.957	1.109	0.990	5/5
APFL, tuned on development [Deng et al. 2020]	0.893	0.966	0.994	1.016	0.960	0.966	5/5
Ditto, tuned on development [Li et al. 2021a]	0.878	**0.893**	0.974	0.982	**0.866**	0.919	3/5
pFedMe, tuned on development [T. Dinh et al. 2020]	1.244	1.410	1.137	1.169	1.494	1.291	5/5
Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018]	0.894	0.940	1.002	1.095	0.894	0.965	5/5
Best of the six, oracle per network⋆	0.878	0.877	0.970	0.925	0.864	0.903	2/5

7.2 Held-out cancer networks

Networks and protocol. The Cancer Genome Atlas records the institution that contributed each patient's tissue, so each cancer type is a network and each institution with at least five eligible patients a hospital [Cancer Genome Atlas Research Network et al. 2013; Grossman et al. 2016]. Every hospital fits a logistic regression with eight coefficients for death within two years (intercept, age, sex, stage III, stage IV, T3 or T4, nodal involvement, prior malignancy; $\|x\| \leq \sqrt{8}$). Twelve networks were used to design GHB: lung squamous and adenocarcinoma, head and neck, bladder, stomach, liver, clear-cell and papillary kidney, colon, rectum, pancreas and oesophagus. Five others were held out, chosen by a rule on counts alone (every further project with at least four hospitals and fifteen deaths): breast (8 hospitals), mesothelioma (9), ovarian (15), endometrial (15) and uterine carcinosarcoma (5), with 5 to 109 patients per hospital. No component of GHB was chosen on them, but they are not unseen: an earlier configuration of GHB, which borrowed the intercept, had been evaluated on them before its three final choices (not borrowing the intercept, the tight sensitivity, the diagonal rule) were made on the design networks, so its results on them were known when those choices were made. Evaluation is by ten repetitions of stratified five-fold cross-validation. The six literature methods are DP-FedAvg [McMahan et al. 2018] with fine-tuning, with and without secure aggregation [Bonawitz et al. 2017], clustered DP-FedAvg [Ghosh et al. 2020], APFL [Deng et al. 2020], Ditto [Li et al. 2021a] and pFedMe [T. Dinh et al. 2020], each with 12 to 54 configurations; the realistic arm uses the configuration best on the twelve design networks. On those networks GHB reaches 0.911 and 0.892 of the local error at $\epsilon = 1$ and 10, against 0.934 and 0.914 for the best literature method and 0.927 and 0.941 for GHB-lite.

Results. GHB has the lowest mean error at both budgets among the methods a consortium could tune (Table 12): 0.925 of the local error at $\epsilon = 1$, against 0.976 for DP-FedAvg [McMahan et al. 2018]

with fine-tuning, and 0.909 at $\epsilon = 10$, against 0.919 for Ditto [Li et al. 2021a]. Tuned on the test folds, which no hospital can do, every literature method is still behind GHB at $\epsilon = 1$ (best 0.939, and 0.933 for the best of the six chosen afresh for each network); at $\epsilon = 10$ the best of them is level with GHB (Ditto [Li et al. 2021a], 0.911 here, 0.909 for the best family when rerun with fresh privacy noise, Section 7.3), and the best of the six chosen afresh for each network with hindsight is ahead (0.903). At $\epsilon = 10$ GHB beats Ditto [Li et al. 2021a] in only 3 of the five networks, losing on the two with the fewest patients. Tuning GHB like its competitors, over 36 configurations, would improve it only slightly (0.920 and 0.900; supplement), and tuning both sides the same way leaves GHB better in 19 of 20 comparisons. Five networks are a small sample: at $\epsilon = 1$ GHB beat the best realistic method on all five, which gives a one-sided Wilcoxon signed-rank p of 0.031 [Wilcoxon 1945], the floor five networks allow, and that does not survive correction for multiplicity (Holm [Holm 1979] $p = 0.125$). We therefore added a second held-out set.

A second held-out set. Every further TCGA project had been used, and no other program of the NCI Genomic Data Commons records the institution, so the new networks come from the Clinical Proteomic Tumor Analysis Consortium (CPTAC) [Edwards et al. 2015], chosen by the same rule on counts and run once, after the networks, the protocol and the tests had been committed (supplement). CPTAC does not release the contributing institution, and its cases were collected by centres in each country, so the patient's country of birth stands in for the hospital, a proxy that can merge centres within a country. Three cohorts qualify: pancreatic adenocarcinoma, cutaneous melanoma and lung adenocarcinoma, with five hospitals each (332 patients, 5 to 50 per hospital). Outcome, features and protocol are those above; GHB fits the three networks together, and the literature keeps the configurations chosen on the TCGA design networks. No configuration of GHB had been run on these data. At $\epsilon = 1$ GHB again has the lowest mean error among the methods a consortium could tune (Table 13: 0.902, against 0.905 for DP-FedAvg [McMahan et al. 2018] with fine-tuning), level with the best of them tuned on the test folds (0.902) and slightly behind the best chosen per network with hindsight (0.899). At $\epsilon = 10$ it does not lead: 3 of the six realistic methods are more accurate on average, Ditto [Li et al. 2021a] the most (0.898 against 0.905).

All eight held-out networks. Pooling both sets gives eight networks (Table 21). The decision to add CPTAC was taken after the five TCGA networks had fallen short of significance; the networks, the pooling and the tests were then fixed before any CPTAC result, but the Holm correction does not account for that first look, so a borderline result here should be read with this in mind. GHB beat local estimation under the default prior on all eight at both budgets, which survives the correction over the four primary tests (Holm $p = 0.016$ and 0.016). Against the strongest method a consortium could tune, it was better on 7 of the 8 networks at $\epsilon = 1$ (mean difference -0.033, 95% interval [-0.076, +0.009]; $p = 0.055$, Holm $p = 0.109$) and on 4 of the 8 at $\epsilon = 10$ (-0.004, interval [-0.023, +0.015]). Against the best of the six chosen per network with hindsight, the mean differences are -0.004 and +0.013. GHB's gain over a hospital alone is therefore established on held-out networks; its lead over the tuned literature is likely at $\epsilon = 1$ but not established, and absent at $\epsilon = 10$, where GHB is level with the best federated and personalized methods.

Who gains, and from whom. On the five TCGA networks most hospitals are tiny: 32 of the 52 hold 5 to 15 eligible patients, and they gain the most (Table 14). At $\epsilon = 10$ their error falls to 0.916 of what they reach alone on average, against 0.971 and 0.967 for hospitals of 16 to 40 and over 40 patients, as Equation (3) predicts for a fixed number of virtual patients. Their partners are just as small; what helps them is the sum. A local model with a better prior, which borrows nothing, recovers only part of the gain (0.970 for the smallest hospitals). Fitted alone, without the other cancer types, the smallest hospitals keep their gain at $\epsilon = 1$ (0.931) but lose part of it at $\epsilon = 10$

Table 13. Three CPTAC networks, a second held-out set (five hospitals each; the hospital is the patient's country of birth): held-out log-loss relative to local estimation, as in Table 12, with the literature's configurations from the TCGA design networks. Bold: the lowest per network among methods that use other hospitals, oracle excluded. Wins: networks in which GHB is more accurate. ‡: no other hospital. ★: best configuration of the best of the six methods on each network's own held-out folds.

Method	Pancreas	Melanoma	Lung adeno.	Mean	Wins
$\epsilon = 1$					
GHB, no tuning	0.849	**0.963**	**0.895**	**0.902**	
GHB-lite (evidence prior), no tuning	0.846	0.982	0.927	0.918	2/3
Local, evidence prior‡	0.847	0.981	0.926	0.918	2/3
DP-FedAvg + FT, tuned on development [McMahan et al. 2018]	**0.823**	0.980	0.913	0.905	2/3
DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018]	0.840	1.009	0.934	0.928	2/3
APFL, tuned on development [Deng et al. 2020]	0.832	0.997	0.937	0.922	2/3
Ditto, tuned on development [Li et al. 2021a]	0.899	1.027	0.962	0.963	3/3
pFedMe, tuned on development [T. Dinh et al. 2020]	13.1	16.7	16.2	15.3	3/3
Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018]	0.828	0.993	0.905	0.908	2/3
Best of the six, oracle per network★	0.813	0.980	0.905	0.899	2/3
$\epsilon = 10$					
GHB, no tuning	0.854	0.962	0.898	0.905	
GHB-lite (evidence prior), no tuning	0.844	0.985	0.931	0.920	2/3
Local, evidence prior‡	0.847	0.981	0.926	0.918	2/3
DP-FedAvg + FT, tuned on development [McMahan et al. 2018]	0.827	0.973	0.910	0.903	2/3
DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018]	0.859	0.975	0.919	0.918	3/3
APFL, tuned on development [Deng et al. 2020]	0.834	0.977	0.909	0.907	2/3
Ditto, tuned on development [Li et al. 2021a]	0.862	**0.941**	**0.893**	**0.898**	1/3
pFedMe, tuned on development [T. Dinh et al. 2020]	1.347	1.504	1.539	1.464	3/3
Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018]	**0.823**	0.976	0.913	0.904	2/3
Best of the six, oracle per network★	0.808	0.937	0.893	0.879	0/3

Table 14. Who gains on the five held-out TCGA networks: held-out log-loss relative to local estimation, mean over the hospitals of each size (ten repetitions). Local, evidence prior: a hospital alone with the evidence-weighted default prior, which borrows nothing.

				GHB, networks together		GHB, each network alone	
Patients per hospital	Hospitals	Patients	Local, evid. prior	$\epsilon = 1$	$\epsilon = 10$	$\epsilon = 1$	$\epsilon = 10$
5 to 15	32	268	0.970	0.936	0.916	0.931	0.937
16 to 40	12	325	1.001	0.977	0.971	0.980	0.962
over 40	8	466	0.997	0.977	0.967	0.984	0.968

(0.937), while hospitals of 16 to 40 patients do slightly better alone (0.962 against 0.971): under weak privacy, borrowing across related networks helps the smallest hospitals most; under strong privacy it makes little difference to them. These are means over few hospitals (32, 12 and 8), to be read as a pattern, not a precise estimate.

Which components matter on real networks. Table 15 removes one component at a time on the twelve design networks, with the same folds and privacy noise, and then varies each setting. Four components carry most of the gain: not borrowing the intercept (0.929 and 0.906 with it borrowed, against 0.911 and 0.892), the reverse of the simulations, whose hospitals share their base rates; borrowing across cancer types (0.919 and 0.904 without it); the public rule for the information matrix, which matters at both ends (the full matrix is worse at $\epsilon = 1$, 0.917, the diagonal at $\epsilon = 10$, 0.907); and propagating the privacy covariance under strong privacy (0.917 without it at $\epsilon = 1$, worse on 11 of twelve networks). The non-borrowing components cost a little at $\epsilon = 10$ (0.886 without them) and the leave-one-out step a little at both budgets, and are kept because they protect atypical hospitals (Section 6.4) and because Proposition 4.1 requires an independent estimate. The settings hardly matter (0.910 to 0.915 at $\epsilon = 1$, 0.888 to 0.906 at $\epsilon = 10$), except the grid of default

Table 15. Ablation and sensitivity of GHB on the twelve TCGA design networks (ten repetitions of five-fold cross-validation, the same folds and privacy noise for every variant): mean held-out log-loss relative to local estimation, and the number of networks in which GHB as released is more accurate than the variant. Top: one component removed or replaced. Bottom: one setting changed from its released value (scale of τ 0.5, non-borrowing weight $\frac{1}{3}$, information share $\frac{1}{2}$, $Q = 12$, default scales $\{0.5, 1, 2.5\}$). Bold: the lowest per block.

	$\epsilon = 1$		$\epsilon = 10$	
Variant	Mean	GHB better	Mean	GHB better
GHB, as released	**0.911**		**0.892**	
without borrowing across networks	0.919	10/12	0.904	11/12
without the non-borrowing components	0.913	3/12	**0.886**	1/12
without propagating the privacy covariance	0.917	11/12	0.893	8/12
without removing the hospital's own data	**0.909**	5/12	0.891	5/12
intercept borrowed like the slopes	0.929	10/12	0.906	11/12
loose sensitivity of the information release	0.910	4/12	0.893	11/12
full information matrix always	0.917	9/12	0.892	0/12
diagonal information always	0.911	2/12	0.907	9/12
Sensitivity: one setting changed				
scale of τ 0.1	0.912	8/12	0.890	4/12
scale of τ 0.25	0.912	7/12	0.890	4/12
scale of τ 1	0.911	8/12	0.894	10/12
scale of τ 2	0.912	8/12	0.897	11/12
non-borrowing weight 0.1	0.912	3/12	0.888	1/12
non-borrowing weight 1	0.912	9/12	0.898	11/12
non-borrowing weight 3	0.915	9/12	0.906	11/12
information share 0.25	0.910	4/12	0.893	10/12
information share 0.75	0.914	9/12	0.895	10/12
$Q = 1$	0.912	7/12	0.890	3/12
$Q = 3$	0.912	6/12	0.891	1/12
$Q = 6$	0.911	5/12	0.892	7/12
$Q = 24$	0.911	6/12	0.891	2/12
default scales {2.5}	0.963	11/12	0.921	11/12
default scales {1}	**0.905**	4/12	**0.887**	3/12
default scales {0.5, 1, 2.5, 5}	0.927	11/12	0.902	11/12

prior scales, which averaging over $\{0.5, 1, 2.5\}$ protects against a bad choice (a single scale of 2.5 reaches 0.963); a single scale of 1 would have been slightly better on these networks (0.905 and 0.887), but choosing it would be tuning on them.

7.3 When does GHB lead?

FLamby, below, is where GHB stops leading, and its networks differ from the cancer networks in three ways: fewer hospitals, larger heads, and a single network fitted alone. To find out which matters, we changed one feature at a time on the 17 TCGA networks and reran every method under the new condition, with the literature re-tuned on the twelve design networks under that same condition (`simulations/stress_tcga.py`; same folds, ten repetitions). Each network was cut to 6 or 4 hospitals drawn at random (uterine carcinosarcoma, with 5, kept whole at 6); the head was widened to 14 or 40 coefficients with pure-noise features, standardized like real ones; each network was fitted alone, without borrowing across cancer types; and all three were combined in the shape of Fed-Heart-Disease. GHB's privacy noise is that of Table 12, so the standard condition reproduces GHB's held-out results exactly; the literature and GHB-lite are rerun with fresh noise, so their standard row differs from that table by noise (Table 16).

Fewer hospitals and larger heads do not remove GHB's lead. With 4 hospitals per network it still reaches 0.927 and 0.925 at $\epsilon = 1$ and 10, against 0.975 and 0.931 for the best realistic literature method, and with 40 coefficients the literature degrades far more than GHB (1.073 against 0.853 at $\epsilon = 1$), because its privacy noise accumulates over its rounds while GHB's two releases fall

Table 16. When GHB leads: mean held-out log-loss relative to local estimation over the five held-out TCGA networks, under one change at a time (ten repetitions). Every method is rerun under each condition; the literature is re-tuned on the twelve design networks under the same condition (realistic: best of the six families on those networks) or on the test folds (oracle: best of the six families). Bold: the best method a consortium could run (local and oracle excluded).

	$\epsilon = 1$					$\epsilon = 10$				
Condition	GHB	GHB-lite	Local, evid. prior	Best lit., realistic	Best lit., oracle*	GHB	GHB-lite	Local, evid. prior	Best lit., realistic	Best lit., oracle*
Standard (as in Table 12)	**0.925**	0.934	0.933	0.971	0.933	**0.909**	0.974	0.933	0.919	0.909
Each network fitted alone	**0.933**	**0.933**	0.933	0.971	0.933	0.930	0.971	0.933	**0.919**	0.909
6 hospitals per network	**0.919**	0.930	0.930	0.977	0.931	**0.914**	0.955	0.930	0.925	0.911
4 hospitals per network	**0.927**	0.935	0.935	0.975	0.943	**0.925**	0.951	0.935	0.931	0.923
14 coefficients	**0.861**	0.876	0.875	0.910	0.878	**0.855**	0.924	0.875	0.869	0.851
40 coefficients	**0.853**	0.871	0.870	1.073	1.058	**0.849**	0.930	0.870	0.886	0.861
4 hospitals, 14 coefficients, alone	**0.879**	**0.879**	0.879	0.923	0.887	**0.872**	0.890	0.879	0.893	0.847

back to the diagonal. What removes the lead is fitting a network alone. Without borrowing across cancer types, GHB only matches a well-regularized hospital at $\epsilon = 1$ (0.933 against 0.933) and falls behind Ditto [Li et al. 2021a] at $\epsilon = 10$ (0.930 against 0.919). A single network of a few hundred patients gives the joint releases too little to beat a method that tunes how much each hospital borrows; several related networks fitted together give them enough. In the combined, FLamby-shaped condition GHB is again level with a regularized hospital at $\epsilon = 1$, ahead of the literature a consortium could tune at $\epsilon = 10$ on average (0.872 against 0.893) but in only 2 of five networks, and behind the literature tuned on the test folds (0.847). GHB's lead is therefore a lead of networks fitted together, which the cancer networks allowed and FLamby does not.

7.4 A public benchmark: FLamby

The cancer networks were built by us from one source. FLamby [Ogier du Terrail et al. 2022] is the standard cross-silo healthcare benchmark, built by others, with natural hospital splits and fixed train and test sets. *Fed-Heart-Disease* has the four UCI hospitals with FLamby's 13 features and a logistic head, scored by accuracy and log-loss; *Fed-TCGA-BRCA* has six regions of breast-cancer patients with 39 features and survival, modeled as an exponential (Poisson) regression and scored by concordance. GHB carries over unchanged, since the survival likelihood is a sum over patients: the per-patient score is $x(d - t)$ and the information $t\,xx^\top$, and the literature methods clip each gradient to the same bound. Each method runs 20 times; the realistic arm uses the configurations fixed on the design networks of Section 7.2. FLamby was among the data used to design GHB, so these results show what it does there, not a held-out test.

Results. GHB improves on a hospital alone on Fed-Heart-Disease and Fed-TCGA-BRCA, but leads only in some settings (Tables 17 and 18). On Fed-TCGA-BRCA it is the most accurate method at $\epsilon = 1$ by a wide margin (concordance 0.785, against 0.747 for the best literature method), level with the best without privacy (0.825 against 0.829 for pFedMe [T. Dinh et al. 2020]), and behind DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation at $\epsilon = 10$ (0.813 against 0.801). On Fed-Heart-Disease it has the lowest log-loss of the federated and personalized methods without privacy (0.415), lies between DP-FedAvg with fine-tuning and secure aggregation (0.419) and Ditto [Li et al. 2021a] (0.429) at $\epsilon = 10$ (0.427), and is slightly behind Ditto at $\epsilon = 1$ (0.443 against 0.430); its accuracy is below DP-FedAvg with fine-tuning and secure aggregation throughout. GHB-lite has a lower log-loss than GHB on this dataset under weak privacy, though not a higher accuracy. Section 7.3 explains the difference: each FLamby dataset is a single network, and fitted alone GHB's

Table 17. FLamby [Ogier du Terrail et al. 2022], Fed-Heart-Disease (four hospitals): accuracy (FLamby's metric, higher is better) and held-out log-loss (lower is better), mean over hospitals and 20 runs. Literature methods use the configurations frozen on the development networks of Section 7.2. Bold: the best per column among methods that use other hospitals, oracle and the non-private pooled model excluded. †: not private. ★: the best of the six methods on each hospital's test set, which no hospital can do.

	$\epsilon = 1$		$\epsilon = 10$		no privacy	
Method	Accuracy	Log-loss	Accuracy	Log-loss	Accuracy	Log-loss
Local	0.788	0.470	0.788	0.470	0.788	0.470
Local, evidence prior	0.788	0.447	0.788	0.447	0.788	0.447
Pooled, no DP†	0.729	0.568	0.729	0.568	0.729	0.568
GHB, no tuning	0.801	0.443	0.805	0.427	0.821	0.415
GHB-lite (evidence prior), no tuning	0.788	0.446	0.797	**0.416**	0.809	**0.403**
DP-FedAvg + FT, tuned on development [McMahan et al. 2018]	0.810	0.485	0.811	0.472	0.810	0.471
DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018]	**0.816**	0.477	**0.828**	0.419	**0.823**	0.418
APFL, tuned on development [Deng et al. 2020]	0.810	0.500	0.815	0.456	0.818	0.456
Ditto, tuned on development [Li et al. 2021a]	0.790	**0.430**	0.790	0.429	0.791	0.429
pFedMe, tuned on development [T. Dinh et al. 2020]	0.550	10.696	0.701	0.818	0.785	0.453
Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018]	0.811	0.485	0.810	0.482	0.810	0.481
Best of the six, oracle★	0.821	0.419	0.834	0.405	0.848	0.399

Table 18. FLamby [Ogier du Terrail et al. 2022], Fed-TCGA-BRCA (six regions): concordance index (FLamby's metric, higher is better) and held-out Poisson deviance (lower is better) of an exponential survival model, mean over regions and 20 runs. Conventions as in Table 17. d: the method diverged; with its privacy noise correctly calibrated, pFedMe's coefficients at $\epsilon = 1$ are so large that its predicted event rates reach the numerical cap e^{50}, and its deviance is of order 10^{20}.

	$\epsilon = 1$		$\epsilon = 10$		no privacy	
Method	C-index	Deviance	C-index	Deviance	C-index	Deviance
Local	0.738	0.648	0.738	0.648	0.738	0.648
Local, evidence prior	0.733	0.479	0.733	0.479	0.733	0.479
Pooled, no DP†	0.808	0.421	0.808	0.421	0.808	0.421
GHB, no tuning	**0.785**	**0.477**	0.801	0.458	0.825	0.446
DP-FedAvg + FT, tuned on development [McMahan et al. 2018]	0.725	0.500	0.771	0.491	0.776	0.492
DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018]	0.747	0.496	**0.813**	**0.440**	0.811	0.441
APFL, tuned on development [Deng et al. 2020]	0.729	0.500	0.774	0.489	0.791	0.486
Ditto, tuned on development [Li et al. 2021a]	0.735	0.528	0.752	0.525	0.753	0.525
pFedMe, tuned on development [T. Dinh et al. 2020]	0.571	$> 10^{3d}$	0.610	5.748	**0.829**	**0.420**
Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018]	0.720	0.501	0.754	0.498	0.753	0.498
Best of the six, oracle★	0.775	0.486	0.814	0.414	0.836	0.395

joint releases carry too little to beat a well-tuned personalized method under weak privacy, while fewer hospitals or a larger head alone do not remove its lead. FLamby also standardizes features within each hospital, which makes GHB-lite's public feature distribution nearly exact and explains why GHB-lite's log-loss is lower there.

7.5 Images: a frozen public encoder

Many small clinical problems are images. GHB extends to them without changing its privacy argument, because everything new is public or local. A frozen public encoder, FLamby's EfficientNet-B0 with ImageNet weights [Ogier du Terrail et al. 2022; Tan and Le 2019], embeds each image, and the features are projected on the first 32 principal components of public unlabeled images. Each hospital fits one softmax head. The network makes one joint release, the secure sum of every patient's score (sensitivity $2\sqrt{2}\,G$ for the softmax), with a small share of the budget for the class counts that start the iteration; the curvature comes from the public images, which needs features and the model but no labels. Every method, GHB and the literature alike, predicts with the logit

Table 19. Images: balanced accuracy (higher is better), mean over hospitals and repetitions. ISIC 2019: the data on which the image version was designed (six centers, eight classes, 10 repetitions). Camelyon17: held-out evaluation (five hospitals, binary, 200 training patches each, 10 repetitions). Every method uses the same frozen encoder, public projection and logit adjustment. Literature methods use the configurations frozen on the development networks of Section 7.2. Bold: the best per column among methods that use other hospitals, oracle excluded. †: not private. ⋆: the best of the six methods on each test set, which no hospital can do.

	ISIC 2019 (design)			Camelyon17 (held out)		
Method	$\epsilon = 1$	$\epsilon = 10$	no privacy	$\epsilon = 1$	$\epsilon = 10$	no privacy
Local	0.533	0.533	0.533	0.885	0.885	0.885
Local, evidence prior	0.516	0.516	0.516	0.893	0.893	0.893
Pooled, no DP†	0.523	0.523	0.523	0.887	0.887	0.887
GHB, no tuning	**0.535**	**0.583**	**0.583**	0.893	**0.896**	**0.896**
DP-FedAvg + FT, tuned on development [McMahan et al. 2018]	0.387	0.399	0.399	**0.894**	0.894	0.895
DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018]	0.396	0.409	0.411	0.889	0.890	0.893
APFL, tuned on development [Deng et al. 2020]	0.392	0.399	0.399	0.889	0.895	**0.896**
Ditto, tuned on development [Li et al. 2021a]	0.511	0.510	0.510	0.893	0.894	0.894
pFedMe, tuned on development [T. Dinh et al. 2020]	0.239	0.338	0.569	0.620	0.772	0.894
Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018]	0.388	0.384	0.380	0.891	0.895	0.895
Best of the six, oracle⋆	0.534	0.534	0.578	0.898	0.899	0.900

adjustment of [Menon et al. 2021] by the class mix that the model implies on the hospital's own training images, restricted to the classes it has seen.

The image version was designed on FLamby's Fed-ISIC2019 [Combalia et al. 2019; Ogier du Terrail et al. 2022] (six centers, eight dermoscopic classes, the ISIC 2019 Test images as public data), where with the tabular head on a random projection every method that borrows had stayed at the local level. With the image version, GHB reaches a balanced accuracy of 0.583 at $\epsilon = 10$, against 0.533 for each hospital alone and 0.510 for the best literature method (Ditto), and is level with a hospital alone at $\epsilon = 1$ (0.535).

It was then evaluated, unchanged, on Camelyon17 [Bándi et al. 2019; Koh et al. 2021]: lymph-node histopathology from five Dutch hospitals, tumor or normal tissue in a 96-pixel patch, rebuilt from the organizers' slides with the WILDS preprocessing code (matching the released Camelyon17-WILDS within 0.9%). Each hospital trains on 200 patches and is tested on 1,000; the public images are 2,000 unlabeled patches from other patients; 10 repetitions. At $\epsilon = 10$ GHB is the most accurate method a consortium could run (Table 19): 0.896, against 0.885 for a hospital alone and 0.895 for the best literature method (APFL). Its lead over a hospital alone is statistically clear (Holm $p \leq 0.010$; Table 21), its lead over the literature established for only 3 of 6 methods. The task is easy for a frozen encoder, and most of the gain is regularization (the local model with the evidence-weighted prior reaches 0.893); at $\epsilon = 1$ DP-FedAvg [McMahan et al. 2018] with fine-tuning is slightly above GHB (0.894 against 0.893), and the best literature method tuned on the test patches is above it at both budgets (0.898 and 0.899). The privacy unit is a patch, not a patient.

7.6 What the releases reveal about patients

Differential privacy is a bound on re-identification, and it can be read as one. GHB's two releases form a Gaussian mechanism with parameter $\mu = \sqrt{2\rho}$ in the sense of Gaussian differential privacy [Dong et al. 2022]: whatever the attacker knows, no test can tell whether a given patient's record was used, against any replacement, with a true-positive rate above $\Phi(\Phi^{-1}(\alpha) + \mu)$ at false-positive rate α. Table 20 gives what this means. At $\epsilon = 1$ no attacker can gain more than 8.1% in true-positive minus false-positive rate, and at a 1% false-positive rate it detects at most 1.7% of members. At $\epsilon = 10$ the guarantee is far weaker, up to 62.1%, which is why weak privacy should be chosen knowingly. These bounds hold against observers outside the network; a participating hospital,

Table 20. Re-identification risk of GHB's two releases. Worst case: the largest advantage (true-positive minus false-positive rate) of any test of whether a patient's record was used, and its largest true-positive rate at a 1% false-positive rate, from the Gaussian differential privacy of the releases. Realized: the same advantage for each of the 3,390 real TCGA patients, with their actual contributions against the most different patient of their network (median and maximum); it is the power of the optimal likelihood-ratio attack by an attacker who knows every other record, which simulation confirms. Attack: AUC of a loss-threshold membership-inference attack against GHB's released network estimate, mean over networks and repetitions (0.5: no information).

	Worst case		Realized, real patients		Attack
Budget	Advantage	TPR at 1% FPR	Median	Maximum	AUC
$\epsilon = 1$	8.1%	1.7%	2.8%	3.7%	0.50
$\epsilon = 10$	62.1%	28.6%	23.6%	30.8%	0.51

which knows its own noise share, faces a budget at most 1.25 times larger in networks of five or more hospitals (Proposition 5.1), and an advantage of at most 9.1% and 67.5%. The bound assumes a patient whose contribution reaches the sensitivity limit; computed with each of the 3,390 real TCGA patients' actual contributions, against the most different patient of the same network, the attainable advantage is at most 3.7% at $\epsilon = 1$ and 30.8% at $\epsilon = 10$.

These realized values are not only a bound: they are the power of the strongest possible attack, the likelihood-ratio test of an attacker who knows every other record and sees both raw releases. Simulating that attack for 300 real patients gives an advantage of 2.7% at $\epsilon = 1$ and 22.9% at $\epsilon = 10$, matching the calculation, and detects 1.2% and 4.2% of members at a 1% false-positive rate. A weaker but common attack, thresholding each patient's loss under the released network estimate [Shokri et al. 2017], learns nothing (AUC 0.50 and 0.51, where 0.5 means no information); it also learns little from the pooled model released without any privacy (0.51), because low-dimensional network summaries leak little in practice, which is why the guarantee, not this attack, is the measure of protection. Against a hospital's own model released without privacy, as federated or meta-analytic sharing would, the same attack reaches 0.62 for hospitals of 5 to 15 patients (supplement): the smallest hospitals' shared models leak most, and GHB never shares them.

7.7 What the real data say

Four findings hold across the real networks. Under strong privacy, per-hospital releases carry too little to borrow, and GHB-lite does no better than a well-regularized hospital. Releasing only network sums of the hospitals' real scores and information is what lets collaboration pay: on the five held-out TCGA networks GHB had a lower mean error than every literature method a consortium could tune, at both budgets; on the three CPTAC networks it did so at $\epsilon = 1$ but not at $\epsilon = 10$, and on all eight it improved on a hospital alone in every network. The gain grows with what the network can share: it comes from fitting several related networks together, survives fewer hospitals and larger heads, and disappears for a single network fitted alone (Section 7.3), which is why it is mixed on FLamby, where Ditto [Li et al. 2021a], DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation or GHB-lite are sometimes ahead, and small on images. And GHB is safe on average: in every study and budget its mean error was at most that of a hospital alone with the default prior, which several tuned literature methods cannot say. GHB leads the tuned literature under strong privacy on most held-out networks, is level with it under weak privacy, and is competitive elsewhere; it does not dominate the literature, and we do not claim that it does.

Table 21. Statistical tests on the confirmatory data (supplement). Mean difference: GHB minus the comparator, relative log-loss on the held-out networks (negative favours GHB), balanced accuracy on Camelyon17 (positive favours GHB), with a 95% t interval. Better: networks or repetitions where GHB is better. p: one-sided exact Wilcoxon signed-rank test; Holm: adjusted within each family (held-out networks: four primary tests per set; Camelyon17: the ten tests of the four claims tested on it). Bold: established at 0.05 after adjustment. Realistic: tuned on other networks; oracle: tuned on the test folds, reported but not tested. On the held-out networks the realistic comparator is the method with the lowest mean at that budget; the eight-network family (TCGA and CPTAC) is primary. Camelyon17 at $\epsilon = 1$ tests non-inferiority with a margin of 0.005.

Data	ε	GHB against	Mean difference [95% CI]	Better	p	Holm p
All held-out (8 networks)	1	Local	−0.084 [−0.122, −0.046]	8 of 8	0.004	**0.016**
All held-out (8 networks)	1	DP-FedAvg + FT [McMahan et al. 2018] (realistic)	−0.033 [−0.076, +0.009]	7 of 8	0.055	0.109
All held-out (8 networks)	1	DP-FedAvg + FT [McMahan et al. 2018] (oracle)	−0.009 [−0.025, +0.007]	7 of 8	not tested	not tested
All held-out (8 networks)	1	Best per network	−0.004 [−0.019, +0.011]	6 of 8	not tested	not tested
All held-out (8 networks)	10	Local	−0.093 [−0.129, −0.057]	8 of 8	0.004	**0.016**
All held-out (8 networks)	10	Ditto [Li et al. 2021a] (realistic)	−0.004 [−0.023, +0.015]	4 of 8	0.422	0.422
All held-out (8 networks)	10	Ditto [Li et al. 2021a] (oracle)	+0.007 [−0.010, +0.024]	3 of 8	not tested	not tested
All held-out (8 networks)	10	Best per network	+0.013 [−0.004, +0.030]	2 of 8	not tested	not tested
TCGA held-out (5 networks)	1	Local	−0.075 [−0.127, −0.023]	5 of 5	0.031	0.125
TCGA held-out (5 networks)	1	DP-FedAvg + FT [McMahan et al. 2018] (realistic)	−0.052 [−0.121, +0.018]	5 of 5	0.031	0.125
TCGA held-out (5 networks)	1	DP-FedAvg + FT [McMahan et al. 2018] (oracle)	−0.014 [−0.024, −0.004]	5 of 5	not tested	not tested
TCGA held-out (5 networks)	10	Local	−0.091 [−0.143, −0.040]	5 of 5	0.031	0.125
TCGA held-out (5 networks)	10	Ditto [Li et al. 2021a] (realistic)	−0.010 [−0.043, +0.023]	3 of 5	0.312	0.312
TCGA held-out (5 networks)	10	Ditto [Li et al. 2021a] (oracle)	−0.003 [−0.023, +0.017]	3 of 5	not tested	not tested
Camelyon17 (10 repetitions)	10	Local	+0.011 [+0.009, +0.013]	10 of 10	<0.001	**0.010**
Camelyon17 (10 repetitions)	10	DP-FedAvg + FT [McMahan et al. 2018] (realistic)	+0.002 [−0.001, +0.005]	7 of 10	0.116	0.349
Camelyon17 (10 repetitions)	10	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018] (realistic)	+0.006 [+0.004, +0.009]	9 of 10	0.002	**0.010**
Camelyon17 (10 repetitions)	10	APFL [Deng et al. 2020] (realistic)	+0.002 [−0.001, +0.004]	7 of 10	0.116	0.349
Camelyon17 (10 repetitions)	10	Ditto [Li et al. 2021a] (realistic)	+0.002 [+0.001, +0.003]	9 of 10	0.005	**0.020**
Camelyon17 (10 repetitions)	10	pFedMe [T. Dinh et al. 2020] (realistic)	+0.124 [+0.101, +0.147]	10 of 10	<0.001	**0.010**
Camelyon17 (10 repetitions)	10	Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] (realistic)	+0.002 [−0.002, +0.005]	6 of 10	0.161	0.349
Camelyon17 (10 repetitions)	1	Local, non-inferiority	+0.008 [+0.005, +0.010]	10 of 10	<0.001	**0.010**
Camelyon17 (10 repetitions)	∞	Local	+0.011 [+0.009, +0.013]	10 of 10	<0.001	**0.010**
Camelyon17 (10 repetitions)	∞	Pooled	+0.009 [+0.007, +0.012]	10 of 10	<0.001	**0.010**

Formal tests (Table 21) separate what is established from what is suggested. Across all 19 real data sets, GHB improved on a hospital alone in 17 at $\epsilon = 1$ and 18 at $\epsilon = 10$ (sign test, Holm $p = 0.001$ and $p =< 0.001$), but beat the strongest realistic competitor of each data set in only 13 and 10, which chance alone could produce (Holm $p = 0.167$ and $p = 0.500$). Ten of these data sets informed GHB's design, so these counts describe rather than confirm. What the evidence establishes is that GHB, untuned, improves on working alone, on all eight held-out networks at both budgets (Holm $p = 0.016$); that it beats the tuned literature is suggested at $\epsilon = 1$, on 7 of the eight held-out networks and part of Camelyon17, but not established, and at $\epsilon = 10$ it does not.

8 DISCUSSION

8.1 When to use GHB, and when not

The evidence supports a plain reading of the theory. Collaboration pays when hospitals are small for their task and differ moderately; then Equation (3) adds tens of virtual patients to every hospital. When they differ strongly, no method gains much, and integrating τ out keeps GHB near local estimation rather than below it. When they barely differ and a trusted pooling environment exists, a pooled model is better, and GHB can run inside it without noise.

Which protocol to run depends on what the network can provide. GHB needs secure summation among hospitals that do not collude, and in exchange calibrates its noise to the whole network and uses the hospitals' real case mix; on the held-out cancer networks it improved on every hospital working alone, led the tuned literature under strong privacy by a margin eight networks do not establish statistically, and was level with it under weak privacy. Its lead comes from fitting several related networks together, for instance one per disease, and survives networks of four hospitals and heads of 40 coefficients (Section 7.3). A single cancer network fitted alone gives it little more than a well-regularized hospital at strong privacy, although on FLamby's survival network GHB still

leads at $\epsilon = 1$, and a consortium able to tune Ditto [Li et al. 2021a] or DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation on comparable networks may do better under weak privacy. GHB-lite needs only honest-but-curious neighbours; it is the better choice when a public feature distribution matches the hospitals' patients and the network is large enough for per-hospital releases to be informative, as in the simulations and, partly, on FLamby. Where hospitals are known to share their base rates, borrowing the intercept, as GHB-lite does, is better. We recommend $\epsilon = 1$, whose guarantee is strong (Section 7.6); $\epsilon = 10$ trades much of it away for accuracy. Before committing, a network can check from public quantities (sizes, budget and a prior guess of τ^2) whether privacy leaves anything to share, with Equations (4) and (3).

8.2 Deploying GHB in a real network

Running GHB asks little of a hospital; the figures below are estimates from our implementation, not measurements in a deployed network. Each one computes two sums over its own patients, the scores and the information of its head (44 numbers for the eight-coefficient cancer head, 352 bytes), adds its share of the noise and pairwise masks, and sends one masked value up a spanning tree of the agreement graph; with the public keys relayed for the masks, that is about 3 kB sent per hospital in the largest network (17 hospitals). Everything after the two releases is local: computing the leave-one-out prior and the evidence-weighted fit takes about 17 ms per hospital on one processor core for the cancer networks. There is no server to procure or trust, and no training rounds to schedule: one release per model fit, repeated only when the model is refitted, with the costs of Section 4.2 adding up under zero-concentrated accounting.

Three practical points remain. Secure summation must tolerate hospitals that drop out mid-protocol, which practical protocols do by secret-sharing each hospital's masks so that the survivors can remove them [Bonawitz et al. 2017]; our implementation simulates secure summation by computing the exact sums, and a real network must adopt such a protocol. The channels can be those that networks already use for distributed analysis, such as the OHDSI network on the OMOP common data model [Hripcsak et al. 2015], DataSHIELD [Gaye et al. 2014] or the Personal Health Train [Beyan et al. 2020; Deist et al. 2020]: they move analyses to the data, and GHB specifies what to compute and release through them. Finally, a network must fix the head, the feature bound G, the budget and the graph before any release, and document the processing as data-protection law requires [European Parliament and Council of the European Union 2016, 2025]. A pilot in a real network is the test this paper does not provide.

8.3 Limits

GHB protects against honest-but-curious hospitals and observers under Proposition 5.1, with a slightly weaker guarantee against a participating hospital, which knows its own noise share; colluders weaken it further, and GHB-lite is the option where collusion cannot be ruled out. A hospital that lies is not excluded: under GHB-lite a large lie inflates $\hat{\tau}^2$ and switches borrowing off (Simulation 12), while GHB, which has no per-hospital release in which to see a lie, relies on each hospital's evidence to stop borrowing from a network that conflicts with it; stronger adversaries need robust aggregation [Blanchard et al. 2017]. Refits compose and must be accounted for [Bun and Steinke 2016; Mironov 2017]. The Gaussian analysis is asymptotic, and GHB's network estimate is one Newton step from the origin, accurate for the clinical heads studied here but not for heads far from it. The evidence has limits too. The design was chosen on development networks, and the held-out evaluation uses eight networks from two sources, three of them with the country of birth standing in for the hospital, and was extended from five to eight networks after the first five fell short of statistical significance, with one outcome and an eight-coefficient head; the margins under weak privacy are small, and FLamby shows that other networks can reverse them. The baselines

were tuned as well as a consortium could, and their oracle arm shows what more tuning buys, but other methods or grids could do better. On images the privacy unit is a patch, not a patient. Nothing here speaks to heads of hundreds of coefficients, other outcomes, much larger networks or adversaries we did not test.

8.4 Open problems

(1) **Private heterogeneity from joint releases.** GHB integrates τ over a fixed prior; GHB-lite learns it from releases that real networks made too noisy. A private estimate of τ^2 from network sums would combine both strengths and let a hospital predict its benefit before joining.

(2) **Topology design.** Since mixing depends on the spectral gap, choosing a few long-range agreements that maximize it is a concrete optimization problem.

(3) **Weaker trust.** Shuffling [Erlingsson et al. 2019] or dropout-tolerant, collusion-detecting summation would keep network-level noise with less trust.

(4) **Incentives.** Large hospitals gain least and contribute most; private data valuation [Ghorbani and Zou 2019] could support fair credit under regulation such as the European Health Data Space [European Parliament and Council of the European Union 2025].

(5) **Shift-aware borrowing.** Separating the transportable part of τ^2 from the site-specific part, with causal invariance [Arjovsky et al. 2019; Castro et al. 2020].

(6) **Group fairness.** Borrowing can import or dilute disparities; Simulation 18 finds both effects small for GHB-lite, but parity of decisions would need further private releases [Salazar et al. 2023].

(7) **Procurement and strata.** Posterior variances could drive joint labeling across a consortium, and Equation (3) applies to small subgroups as well as small sites.

9 CONCLUSION

Networks of small hospitals without a trusted centre are the default form of healthcare, and they are poorly served by methods that assume a server, fit one model or must be tuned on data small hospitals do not have. Graph Hierarchical Borrowing gives each hospital almost the same number of virtual patients, pays the privacy cost once, in two sums over the whole network, runs over whatever agreements exist, and tunes nothing. On eight held-out cancer networks it improved on a hospital alone in every network, at strong and weak privacy, and was more accurate than federated and personalized methods tuned as a consortium could on seven of eight under strong privacy, a lead too small to establish statistically, and level with them under weak privacy; the lead comes from borrowing across related networks, and a single network fitted alone gives it no more than a well-regularized hospital; on smaller benchmarks and on images it is competitive, leading clearly in one setting and by small margins or not at all in others. Collaboration between small hospitals pays under privacy, by percent rather than by multiples, when the noise is calibrated to the network rather than to each hospital. Whether this holds for other outcomes, much larger heads and larger networks, and in a deployed network, is the main open question.

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